

# Additional file 1: Supplementary tables, notes and figures

Ensembles Across Model Families Outperform Any Single Family:  
A Single-Configuration, Leakage-Controlled Benchmark of 31 Molecular Property Models  
Across 44 Datasets

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This file contains the supplementary tables (S1–S16), notes (S1–S6) and figures (S1–S5) referenced in the main text. Notes S1, S2, S5 and S6 hold the detail that the revised main text summarizes; Note S3 reports two sensitivity analyses of the family comparison; Note S4 is the dataset-property meta-analysis. Tables S1–S7, S12 and S16 and Figures S4–S5 are generated from the deposited benchmark artifacts by `portable_colab_qsar_bundle/render_manuscript_assets.py` and `render_latex_tables.py`; Table S8 by `python -m qsarena.reliability_tables` from the reliability-study artifacts; Tables S9–S11, Figures S1–S3 and the text of Note S4 by `qsarena.meta_analysis`, which the same render script runs; and Tables S13–S14 by `reanalysis_coverage_posthoc.py`. They carry the same values as the machine-readable CSV and Markdown files in the repository.

All results derive from a single split and a single seed per dataset; as set out in the Limitations section of the main text, individual entries are therefore provisional and close comparisons between models should not be treated as established.

Table S1: Per-dataset winning model under the ranking metric, with the corresponding cross-validation-selected model and its relative gap to the dataset best (%).

| dataset                            | suite | task - kind    | family                          | model                                           | analysis - analysis_metric_value | cv - selected - model | cv - selected - family        | cv_selected_gap_to_best      |        |
|------------------------------------|-------|----------------|---------------------------------|-------------------------------------------------|----------------------------------|-----------------------|-------------------------------|------------------------------|--------|
| tdc_ames                           | TDC   | classification | Ensemble (stacking / averaging) | Ensemble (OOF Stacking (RidgeCV, 5-fold))       | test_roc_auc                     | 0.883                 | Extra trees                   | Conventional ML              | 1.386  |
| tdc_bbb_martins                    | TDC   | classification | Conventional ML                 | Random forest                                   | test_roc_auc                     | 0.925                 | XGBoost (ADMETboost features) | Conventional ML              | 0.963  |
| tdc_bioavailability_ma             | TDC   | classification | Conventional ML                 | CatBoost                                        | test_roc_auc                     | 0.777                 | XGBoost (ADMETboost features) | Conventional ML              | 7.129  |
| tdc_carcinogens_lagunin            | TDC   | classification | Chemprop v2 GNN                 | Chemprop v2 (AttentiveFP, ensemble=3)           | test_roc_auc                     | 0.929                 | XGBoost (ADMETboost features) | Conventional ML              | 2.237  |
| tdc_clintox                        | TDC   | classification | CFA combinatorial fusion        | CFA (Combinatorial Fusion)                      | test_roc_auc                     | 0.974                 | AdaBoost                      | Conventional ML              | 8.986  |
| tdc_cyp1a2_veith                   | TDC   | classification | Ensemble (stacking / averaging) | Ensemble (Weighted average (inverse OOF error)) | test_roc_auc                     | 0.974                 | SVC                           | Conventional ML              | 1.196  |
| tdc_cyp2c19_veith                  | TDC   | classification | Ensemble (stacking / averaging) | Ensemble (OOF Stacking (RidgeCV, 5-fold))       | test_roc_auc                     | 0.938                 | XGBoost                       | Conventional ML              | 1.338  |
| tdc_cyp2c9_substrate_carbonmangels | TDC   | classification | Chemprop v2 GNN                 | Chemprop v2 (AttentiveFP, ensemble=3)           | test_auprc                       | 0.438                 | SVC                           | Conventional ML              | 31.368 |
| tdc_cyp2c9_veith                   | TDC   | classification | Ensemble (stacking / averaging) | Ensemble (OOF Stacking (RidgeCV, 5-fold))       | test_auprc                       | 0.827                 | AdaBoost                      | Conventional ML              | 14.709 |
| tdc_cyp2d6_substrate_carbonmangels | TDC   | classification | Chemprop v2 GNN                 | Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)      | test_auprc                       | 0.693                 | ChemML MLP (PyTorch)          | Deep tabular NN (ChemML MLP) | 26.129 |
| tdc_cyp2d6_veith                   | TDC   | classification | Ensemble (stacking / averaging) | Ensemble (OOF Stacking (RidgeCV, 5-fold))       | test_auprc                       | 0.763                 | AdaBoost                      | Conventional ML              | 19.417 |
| tdc_cyp3a4_substrate_carbonmangels | TDC   | classification | Conventional ML                 | XGBoost (ADMETboost features)                   | test_auprc                       | 0.729                 | XGBoost (ADMETboost features) | Conventional ML              | 0.000  |
| tdc_cyp3a4_veith                   | TDC   | classification | Ensemble (stacking / averaging) | Ensemble (OOF Stacking (RidgeCV, 5-fold))       | test_auprc                       | 0.905                 | AdaBoost                      | Conventional ML              | 11.894 |

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| dataset                    | suite           | task -<br>kind | family                                | model                                                 | analysis_<br>metric  | analysis_metric_value | cv_<br>selected_<br>model           | cv_<br>selected_<br>family        | cv_selected_gap_to_best |
|----------------------------|-----------------|----------------|---------------------------------------|-------------------------------------------------------|----------------------|-----------------------|-------------------------------------|-----------------------------------|-------------------------|
| tdc_dili                   | TDC             | classification | Ensemble<br>(stacking /<br>averaging) | Ensemble (OOF<br>Stacking (RidgeCV, 5-<br>fold))      | test_<br>roc_<br>auc | 0.933                 | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 1.072                   |
| tdc_herg                   | TDC             | classification | Chemprop v2<br>GNN                    | Chemprop v2<br>(AttentiveFP,<br>ensemble=3)           | test_<br>roc_<br>auc | 0.857                 | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 7.662                   |
| tdc_herg_karim             | TDC             | classification | Ensemble<br>(stacking /<br>averaging) | Ensemble (Weighted<br>average (inverse OOF<br>error)) | test_<br>roc_<br>auc | 0.907                 | XGBoost                             | Conventional<br>ML                | 0.494                   |
| tdc_hia_hou                | TDC             | classification | Conventional<br>ML                    | XGBoost<br>(ADMETboost features)                      | test_<br>roc_<br>auc | 0.994                 | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 0.000                   |
| tdc_pampa_<br>ncats        | TDC             | classification | Uni-Mol (3D<br>pretrained)            | Uni-Mol V1                                            | test_<br>roc_<br>auc | 0.763                 | Random forest                       | Conventional<br>ML                | 10.598                  |
| tdc_pgp_<br>broccatelli    | TDC             | classification | Uni-Mol (3D<br>pretrained)            | Uni-Mol V2 (84m)                                      | test_<br>roc_<br>auc | 0.933                 | Random forest                       | Conventional<br>ML                | 3.065                   |
| tdc_skin_<br>reaction      | TDC             | classification | Conventional<br>ML                    | XGBoost<br>(ADMETboost features)                      | test_<br>roc_<br>auc | 0.667                 | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 0.000                   |
| tdc_tox21                  | TDC             | classification | Chemprop v2<br>GNN                    | Chemprop v2<br>(AttentiveFP,<br>ensemble=3)           | test_<br>roc_<br>auc | 0.597                 | Extra trees                         | Conventional<br>ML                | 8.142                   |
| tdc_toxcast                | TDC             | classification | Conventional<br>ML                    | HistGradientBoosting                                  | test_<br>roc_<br>auc | 0.716                 | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 1.662                   |
| chemml_cep_<br>homo        | ChemML          | regression     | Ensemble<br>(stacking /<br>averaging) | Ensemble (OOF<br>Stacking (RidgeCV, 5-<br>fold))      | test_<br>rmse        | 0.090                 | TabPFNRegressor                     | TabPFN<br>(tabular<br>foundation) | 13.888                  |
| chemml_<br>organic_density | ChemML          | regression     | TabPFN<br>(tabular<br>foundation)     | TabPFNRegressor                                       | test_<br>rmse        | 0.005                 | TabPFNRegressor                     | TabPFN<br>(tabular<br>foundation) | 0.000                   |
| esol_delaney               | Molecule<br>Net | regression     | Ensemble<br>(stacking /<br>averaging) | Ensemble (Weighted<br>average (inverse OOF<br>error)) | test_<br>rmse        | 0.622                 | TabPFNRegressor                     | TabPFN<br>(tabular<br>foundation) | 8.211                   |
| freesolv_sampl             | Molecule<br>Net | regression     | TabPFN<br>(tabular<br>foundation)     | TabPFNRegressor                                       | test_<br>rmse        | 0.933                 | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 121.064                 |
| lipophilicity              | Molecule<br>Net | regression     | Ensemble<br>(stacking /<br>averaging) | Ensemble (OOF<br>Stacking (RidgeCV, 5-<br>fold))      | test_<br>rmse        | 0.507                 | TabPFNRegressor                     | TabPFN<br>(tabular<br>foundation) | 17.351                  |
| poduam_pod_<br>nc_std      | PODUAM          | regression     | Ensemble<br>(stacking /<br>averaging) | Ensemble (OOF<br>Stacking (RidgeCV, 5-<br>fold))      | test_<br>rmse        | 0.694                 | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 5.048                   |

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| dataset                                  | suite   | task -<br>kind | family                                | model                                                 | analysis_<br>metric | analysis_metric_value | cv_<br>selected_<br>model           | cv_<br>selected_<br>family        | cv_selected_gap_to_best |
|------------------------------------------|---------|----------------|---------------------------------------|-------------------------------------------------------|---------------------|-----------------------|-------------------------------------|-----------------------------------|-------------------------|
| poduam_pod_<br>rd_std                    | PODUAM  | regression     | Ensemble<br>(stacking /<br>averaging) | Ensemble (OOF<br>Stacking (RidgeCV, 5-<br>fold))      | test_<br>rmse       | 0.556                 | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 1.824                   |
| polaris_adme_<br>fang_hppb_1             | Polaris | regression     | Ensemble<br>(stacking /<br>averaging) | Ensemble (OOF<br>Stacking (RidgeCV, 5-<br>fold))      | test_<br>rmse       | 0.422                 | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 9.517                   |
| polaris_adme_<br>fang_perm_1             | Polaris | regression     | Ensemble<br>(stacking /<br>averaging) | Ensemble (OOF<br>Stacking (RidgeCV, 5-<br>fold))      | test_<br>rmse       | 0.379                 | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 16.319                  |
| polaris_adme_<br>fang_rclint_1           | Polaris | regression     | Ensemble<br>(stacking /<br>averaging) | Ensemble (OOF<br>Stacking (RidgeCV, 5-<br>fold))      | test_<br>rmse       | 0.484                 | XGBoost                             | Conventional<br>ML                | 11.804                  |
| polaris_adme_<br>fang_rppb_1             | Polaris | regression     | Ensemble<br>(stacking /<br>averaging) | Ensemble (OOF<br>Stacking (RidgeCV, 5-<br>fold))      | test_<br>rmse       | 0.478                 | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 6.138                   |
| polaris_adme_<br>fang_solu_1             | Polaris | regression     | Uni-Mol (3D<br>pretrained)            | Uni-Mol V2 (84m)                                      | test_<br>rmse       | 0.551                 | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 7.782                   |
| tdc_caco2_wang                           | TDC     | regression     | Ensemble<br>(stacking /<br>averaging) | Ensemble (OOF<br>Stacking (RidgeCV, 5-<br>fold))      | test_<br>rmse       | 0.338                 | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 11.813                  |
| tdc_clearance_<br>hepatocyte_az          | TDC     | regression     | MapLight +<br>GNN                     | MapLight + GNN (Cat<br>Boost, Strict Parity)          | test_<br>rmse       | 44.113                | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 6.388                   |
| tdc_clearance_<br>microsome_az           | TDC     | regression     | Uni-Mol (3D<br>pretrained)            | Uni-Mol V2 (84m)                                      | test_<br>rmse       | 33.507                | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 13.113                  |
| tdc_half_life_<br>obach                  | TDC     | regression     | Uni-Mol (3D<br>pretrained)            | Uni-Mol V1                                            | test_<br>rmse       | 19.128                | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 77.574                  |
| tdc_<br>hydrationfreeenergy_<br>freesolv | TDC     | regression     | Ensemble<br>(stacking /<br>averaging) | Ensemble (Weighted<br>average (inverse OOF<br>error)) | test_<br>rmse       | 1.075                 | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 5.821                   |
| tdc_ld50_zhu                             | TDC     | regression     | TabPFN<br>(tabular<br>foundation)     | TabPFNRegressor                                       | test_<br>rmse       | 0.806                 | XGBoost                             | Conventional<br>ML                | 4.826                   |
| tdc_<br>lipophilicity_<br>astrazeneca    | TDC     | regression     | Ensemble<br>(stacking /<br>averaging) | Ensemble (OOF<br>Stacking (RidgeCV, 5-<br>fold))      | test_<br>rmse       | 0.551                 | TabPFNRegressor                     | TabPFN<br>(tabular<br>foundation) | 8.398                   |
| tdc_ppbr_az                              | TDC     | regression     | Ensemble<br>(stacking /<br>averaging) | Ensemble (OOF<br>Stacking (RidgeCV, 5-<br>fold))      | test_<br>rmse       | 10.768                | TabPFNRegressor                     | TabPFN<br>(tabular<br>foundation) | 21.078                  |
| tdc_solubility_<br>aqsoldb               | TDC     | regression     | Ensemble<br>(stacking /<br>averaging) | Ensemble (OOF<br>Stacking (RidgeCV, 5-<br>fold))      | test_<br>rmse       | 0.955                 | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML                | 4.572                   |

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| dataset               | suite | task_<br>kind | family            | model                                        | analysis_<br>metric | analysis_<br>metric_<br>value | cv_<br>selected_<br>model           | cv_<br>selected_<br>family | cv_selected_gap_to_best |
|-----------------------|-------|---------------|-------------------|----------------------------------------------|---------------------|-------------------------------|-------------------------------------|----------------------------|-------------------------|
| tdc_vdss_<br>lombardo | TDC   | regression    | MapLight +<br>GNN | MapLight + GNN (Cat<br>Boost, Strict Parity) | test_<br>rmse       | 4.670                         | XGBoost<br>(ADMETboost<br>features) | Conventional<br>ML         | 69.878                  |

Table S2: Staged component ablation: datasets whose best achievable primary metric improved when each pipeline stage was added, computed from the existing model results.

| stage_order | stage                                       | datasets_evaluated | datasets_improved_vs_previous | improvement_fraction |
|-------------|---------------------------------------------|--------------------|-------------------------------|----------------------|
| 1           | Conventional<br>ML<br>only                  | 44                 | 0                             | 0.000                |
| 2           | + Map<br>Light<br>classic<br>fea-<br>tures  | 44                 | 9                             | 0.205                |
| 3           | + neu-<br>ral/<br>deep<br>back-<br>ends     | 44                 | 26                            | 0.591                |
| 4           | +<br>CFA<br>fusion                          | 44                 | 6                             | 0.136                |
| 5           | Full<br>pipeline<br>incl.<br>ensem-<br>bles | 44                 | 22                            | 0.500                |

Table S3: Feature-family selection summary across all datasets, sorted by per-feature enrichment relative to a uniform-selection baseline.

| Feature family      | Datasets selected | Available (sum) | Selected (sum) | % of selected | Enrichment vs uniform |
|---------------------|-------------------|-----------------|----------------|---------------|-----------------------|
| rdkit               | 40                | 8,261           | 1,362          | 11.22         | 5.16                  |
| erg                 | 38                | 12,552          | 1,146          | 9.44          | 2.86                  |
| avalon              | 43                | 44,384          | 3,603          | 29.69         | 2.54                  |
| maccs               | 38                | 5,962           | 224            | 1.85          | 1.18                  |
| maplight            | 41                | 30,070          | 854            | 7.04          | 0.89                  |
| fcfp6               | 43                | 42,500          | 998            | 8.22          | 0.73                  |
| atom_pair           | 41                | 39,651          | 830            | 6.84          | 0.66                  |
| rdk_path            | 40                | 45,036          | 896            | 7.38          | 0.62                  |
| topological_torsion | 40                | 23,653          | 421            | 3.47          | 0.56                  |
| ecfp6               | 37                | 43,299          | 731            | 6.02          | 0.53                  |
| layered             | 40                | 43,642          | 674            | 5.55          | 0.48                  |
| morgan              | 38                | 40,827          | 398            | 3.28          | 0.31                  |

Table S4: Model coverage: datasets attempted and datasets yielding a valid metric, per model.

| model                                                   | datasets_attempted | datasets_valid |
|---------------------------------------------------------|--------------------|----------------|
| Uni-Mol V2 (310m)                                       | 8                  | 0              |
| Uni-Mol V2 (164m)                                       | 4                  | 0              |
| Uni-Mol V2 (84m)                                        | 44                 | 17             |
| TabPFNClassifier                                        | 22                 | 19             |
| ElasticNetCV                                            | 22                 | 20             |
| LogisticRegression                                      | 22                 | 22             |
| MapLight CatBoost (Strict Parity)                       | 22                 | 22             |
| TabPFNRegressor                                         | 22                 | 22             |
| SVC                                                     | 22                 | 22             |
| SVR                                                     | 22                 | 22             |
| Voting Classifier (KNN, SVM)                            | 22                 | 22             |
| Voting Regressor (KNN, SVM)                             | 22                 | 22             |
| Tabular CNN                                             | 22                 | 22             |
| Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)              | 44                 | 38             |
| Chemprop v2 (AttentiveFP, ensemble=3)                   | 44                 | 38             |
| Chemprop v2 (D-MPNN + Selected descriptors, ensemble=3) | 44                 | 42             |
| Chemprop v2 (D-MPNN, ensemble=3)                        | 44                 | 42             |
| Chemprop v2 (CMPNN, ensemble=3)                         | 44                 | 42             |
| HistGradientBoosting                                    | 44                 | 44             |
| Ensemble (OOF Stacking (RidgeCV, 5-fold))               | 44                 | 44             |
| ChemML MLP (PyTorch)                                    | 44                 | 44             |
| CatBoost                                                | 44                 | 44             |
| CFA (Combinatorial Fusion)                              | 44                 | 44             |
| ChemML MLP (TensorFlow)                                 | 44                 | 44             |
| AdaBoost                                                | 44                 | 44             |
| Extra trees                                             | 44                 | 44             |
| Ensemble (Weighted average (inverse OOF error))         | 44                 | 44             |
| Random forest                                           | 44                 | 44             |
| MapLight + GNN (CatBoost, Strict Parity)                | 44                 | 44             |
| Uni-Mol V1                                              | 44                 | 44             |
| Tabular MLP                                             | 44                 | 44             |
| XGBoost                                                 | 44                 | 44             |
| XGBoost (ADMETboost features)                           | 44                 | 44             |

Table S5: Run-to-run comparison: the canonical NSF ACCESS Jetstream2 A100 benchmark against the earlier consumer-GPU (RTX 4060) run, analysed identically. “Same split” marks datasets where both runs used the identical held-out partition; the remainder were re-split to scaffold splits in the A100 run and are therefore not directly comparable. Change is relative and metric-direction aware (positive favours the A100 run).

| Dataset                            | Task           | Same split | A100 best model                            | A100 value | RTX 4060 best model                      | RTX 4060 value | Change (%) |
|------------------------------------|----------------|------------|--------------------------------------------|------------|------------------------------------------|----------------|------------|
| tdc_ames                           | classification | True       | TabPFNClassifier                           | 0.876      | XGBoost                                  | 0.875          | 0.166      |
| tdc_bbb_martins                    | classification | True       | Random forest                              | 0.925      | Random forest                            | 0.932          | -0.788     |
| tdc_bioavailability_ma             | classification | True       | CatBoost                                   | 0.777      | CatBoost                                 | 0.777          | 0.000      |
| tdc_cyp1a2_veith                   | classification | True       | XGBoost                                    | 0.969      | Uni-Mol V1                               | 0.969          | -0.049     |
| tdc_cyp2c19_veith                  | classification | True       | XGBoost                                    | 0.926      | Uni-Mol V1                               | 0.921          | 0.551      |
| tdc_cyp2c9_substrate_carbonmangels | classification | True       | Chemprop v2 (AttentiveFP, ensemble=3)      | 0.438      | Uni-Mol V1                               | 0.454          | -3.439     |
| tdc_cyp2c9_veith                   | classification | True       | XGBoost                                    | 0.796      | HistGradientBoosting                     | 0.792          | 0.541      |
| tdc_cyp2d6_substrate_carbonmangels | classification | True       | Chemprop v2 (D-MPNN + RDKit2D, ensemble=3) | 0.693      | Random forest                            | 0.665          | 4.143      |
| tdc_cyp2d6_veith                   | classification | True       | XGBoost                                    | 0.729      | XGBoost                                  | 0.722          | 0.904      |
| tdc_cyp3a4_substrate_carbonmangels | classification | True       | XGBoost (ADMETboost features)              | 0.729      | LogisticRegression                       | 0.717          | 1.759      |
| tdc_cyp3a4_veith                   | classification | True       | XGBoost (ADMETboost features)              | 0.883      | XGBoost                                  | 0.883          | 0.002      |
| tdc_dili                           | classification | True       | XGBoost (ADMETboost features)              | 0.923      | Uni-Mol V1                               | 0.914          | 0.903      |
| tdc_herg                           | classification | True       | Chemprop v2 (AttentiveFP, ensemble=3)      | 0.857      | AdaBoost                                 | 0.861          | -0.411     |
| tdc_herg_karim                     | classification | True       | XGBoost                                    | 0.902      | Uni-Mol V1                               | 0.893          | 1.055      |
| tdc_hia_hou                        | classification | True       | XGBoost (ADMETboost features)              | 0.994      | CatBoost                                 | 0.989          | 0.499      |
| tdc_pgp_broccatelli                | classification | True       | Uni-Mol V2 (84m)                           | 0.933      | TabPFNClassifier                         | 0.927          | 0.625      |
| chemml_cep_homo                    | regression     | True       | TabPFNRegressor                            | 0.102      | TabPFNRegressor                          | 0.086          | -19.231    |
| chemml_organic_density             | regression     | True       | TabPFNRegressor                            | 0.005      | TabPFNRegressor                          | 0.005          | 5.907      |
| esol_delaney                       | regression     | True       | MapLight CatBoost (Strict Parity)          | 0.645      | TabPFNRegressor                          | 0.612          | -5.269     |
| freesolv_sampl                     | regression     | True       | TabPFNRegressor                            | 0.933      | Chemprop v2 (AttentiveFP, ensemble=1)    | 1.080          | 13.562     |
| lipophilicity                      | regression     | True       | Chemprop v2 (D-MPNN, ensemble=3)           | 0.551      | Uni-Mol V1                               | 0.592          | 6.932      |
| poduam_pod_nc_std                  | regression     | True       | Random forest                              | 0.720      | Extra trees                              | 0.701          | -2.720     |
| poduam_pod_rd_std                  | regression     | True       | XGBoost (ADMETboost features)              | 0.567      | XGBoost                                  | 0.551          | -2.872     |
| polaris_adme_fang_hppb_1           | regression     | True       | Chemprop v2 (D-MPNN, ensemble=3)           | 0.444      | MapLight + GNN (CatBoost, Strict Parity) | 0.449          | 1.072      |
| polaris_adme_fang_perm_1           | regression     | True       | Uni-Mol V1                                 | 0.399      | Uni-Mol V1                               | 0.399          | -0.040     |
| polaris_adme_fang_rclint_1         | regression     | True       | Chemprop v2 (D-MPNN, ensemble=3)           | 0.517      | Uni-Mol V1                               | 0.512          | -0.968     |
| polaris_adme_fang_rppb_1           | regression     | True       | MapLight + GNN (CatBoost, Strict Parity)   | 0.493      | MapLight + GNN (CatBoost, Strict Parity) | 0.494          | 0.170      |
| polaris_adme_fang_solu_1           | regression     | True       | Uni-Mol V2 (84m)                           | 0.551      | Uni-Mol V1                               | 0.575          | 4.202      |
| tdc_caco2_wang                     | regression     | True       | MapLight CatBoost (Strict Parity)          | 0.350      | MapLight CatBoost (Strict Parity)        | 0.350          | 0.000      |
| tdc_clearance_hepatocyte_az        | regression     | True       | MapLight + GNN (CatBoost, Strict Parity)   | 44.113     | MapLight + GNN (CatBoost, Strict Parity) | 43.976         | -0.311     |
| tdc_clearance_microsome_az         | regression     | True       | Uni-Mol V2 (84m)                           | 33.507     | Uni-Mol V1                               | 35.888         | 6.634      |

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| Dataset                          | Task           | Same split | A100 best model                          | A100 value | RTX 4060 best model                      | RTX 4060 value | Change (%) |
|----------------------------------|----------------|------------|------------------------------------------|------------|------------------------------------------|----------------|------------|
| tdc_half_life_obach              | regression     | True       | Uni-Mol V1                               | 19.128     | Uni-Mol V1                               | 17.222         | -11.067    |
| tdc_ld50_zhu                     | regression     | True       | TabPFNRegressor                          | 0.806      | HistGradientBoosting                     | 0.839          | 3.931      |
| tdc_lipophilicity_astrazeneca    | regression     | True       | Chemprop v2 (D-MPNN, ensemble=3)         | 0.577      | Uni-Mol V1                               | 0.587          | 1.698      |
| tdc_ppbr_az                      | regression     | True       | MapLight + GNN (CatBoost, Strict Parity) | 11.356     | MapLight + GNN (CatBoost, Strict Parity) | 11.444         | 0.762      |
| tdc_solubility_aqsolddb          | regression     | True       | TabPFNRegressor                          | 0.961      | Uni-Mol V1                               | 0.989          | 2.807      |
| tdc_vdss_lombardo                | regression     | True       | MapLight + GNN (CatBoost, Strict Parity) | 4.670      | MapLight + GNN (CatBoost, Strict Parity) | 4.668          | -0.035     |
| tdc_carcinogens_lagunin          | classification | False      | Chemprop v2 (AttentiveFP, ensemble=3)    | 0.929      | AdaBoost                                 | 0.863          | 7.656      |
| tdc_clintox                      | classification | False      | Random forest                            | 0.956      | CatBoost                                 | 0.949          | 0.788      |
| tdc_pampa_ncats                  | classification | False      | Uni-Mol V1                               | 0.763      | Extra trees                              | 0.805          | -5.206     |
| tdc_skin_reaction                | classification | False      | XGBoost (ADMETboost features)            | 0.667      | TabPFNClassifier                         | 0.747          | -10.737    |
| tdc_tox21                        | classification | False      | Chemprop v2 (AttentiveFP, ensemble=3)    | 0.597      | XGBoost                                  | 0.812          | -26.451    |
| tdc_toxcast                      | classification | False      | HistGradientBoosting                     | 0.716      | CatBoost                                 | 0.791          | -9.562     |
| tdc_hydrationfreeenergy_freesolv | regression     | False      | TabPFNRegressor                          | 1.116      | TabPFNRegressor                          | 0.592          | -88.600    |

Table S6: Dataset catalog. “Ranking metric” is the metric used for cross-dataset model ranking (RMSE for regression; the dataset’s designated primary classification metric otherwise); leaderboard comparisons in Table S7 use the leaderboard’s own metric, which differs for several TDC regression tasks.

| Dataset                            | Suite | Task           | Molecules | Train  | Test  | Split      | Target scale | Ranking metric | Best model                                      | Best value | QSARena            |             | Est. rank | Best published | Best published model |
|------------------------------------|-------|----------------|-----------|--------|-------|------------|--------------|----------------|-------------------------------------------------|------------|--------------------|-------------|-----------|----------------|----------------------|
|                                    |       |                |           |        |       |            |              |                |                                                 |            | Leaderboard metric | (lb metric) |           |                |                      |
| tdc_ames                           | TDC   | classification | 7,278     | 5,821  | 1,457 | predefined | raw          | roc_auc        | Ensemble (OOFS<br>tacking<br>(RidgeCV, 5-fold)) | 0.883      | ROC_-<br>AUC       | 0.883       | 2         | 0.912          | QW-MTL               |
| tdc_bbb_martins                    | TDC   | classification | 2,030     | 1,624  | 406   | predefined | raw          | roc_auc        | Random forest                                   | 0.925      | ROC_-<br>AUC       | 0.925       | 2         | 0.941          | MolGPS (3B)          |
| tdc_bioavailability_ma             | TDC   | classification | 640       | 512    | 128   | predefined | raw          | roc_auc        | CatBoost                                        | 0.777      | ROC_-<br>AUC       | 0.777       | 1         | 0.748          | MaxQsaring           |
| tdc_carcinogens_lagunin            | TDC   | classification | 280       | 223    | 57    | scaffold   | raw          | roc_auc        | Chemprop v2 (AttentiveFP, ensemble=3)           | 0.929      | ROC_-<br>AUC       | 0.929       | 1         | 0.848          | FATE-Tox (MTL)       |
| tdc_clintox                        | TDC   | classification | 1,478     | 1,180  | 298   | scaffold   | raw          | roc_auc        | CFA (Combinatorial Fusion)                      | 0.974      | ROC_-<br>AUC       | 0.974       | 3         | 0.996          | PrismNet             |
| tdc_cyp1a2_veith                   | TDC   | classification | 12,579    | 10,061 | 2,518 | scaffold   | raw          | roc_auc        | Ensemble (Weighted average (inverse OOF error)) | 0.974      |                    |             |           |                |                      |
| tdc_cyp2c19_veith                  | TDC   | classification | 12,665    | 10,131 | 2,534 | scaffold   | raw          | roc_auc        | Ensemble (OOFS<br>tacking<br>(RidgeCV, 5-fold)) | 0.938      |                    |             |           |                |                      |
| tdc_cyp2c9_substrate_carbonmangels | TDC   | classification | 669       | 534    | 135   | predefined | raw          | auprc          | Chemprop v2 (AttentiveFP, ensemble=3)           | 0.438      | AUPRC              | 0.438       | 3         | 0.450          | MaxQsaring           |
| tdc_cyp2c9_veith                   | TDC   | classification | 12,092    | 9,673  | 2,419 | predefined | raw          | auprc          | Ensemble (OOFS<br>tacking<br>(RidgeCV, 5-fold)) | 0.827      | AUPRC              | 0.827       | 3         | 0.877          | MaxQsaring           |
| tdc_cyp2d6_substrate_carbonmangels | TDC   | classification | 667       | 532    | 135   | predefined | raw          | auprc          | Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)      | 0.693      | AUPRC              | 0.693       | 4         | 0.766          | MaxQsaring           |
| tdc_cyp2d6_veith                   | TDC   | classification | 13,130    | 10,504 | 2,626 | predefined | raw          | auprc          | Ensemble (OOFS<br>tacking<br>(RidgeCV, 5-fold)) | 0.763      | AUPRC              | 0.763       | 2         | 0.811          | MaxQsaring           |
| tdc_cyp3a4_substrate_carbonmangels | TDC   | classification | 670       | 535    | 135   | predefined | raw          | auprc          | XGBoost (ADMETboost features)                   | 0.729      | ROC_-<br>AUC       | 0.665       | 6         | 0.692          | MolE                 |

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| Dataset                | Suite       | Task           | Molecules | Train  | Test  | Split              | Target scale | Ranking metric | Best model                                      | Best value | QSARena            |             | Est. rank | Best published | Best published model |
|------------------------|-------------|----------------|-----------|--------|-------|--------------------|--------------|----------------|-------------------------------------------------|------------|--------------------|-------------|-----------|----------------|----------------------|
|                        |             |                |           |        |       |                    |              |                |                                                 |            | Leaderboard metric | (lb metric) |           |                |                      |
| tdc_cyp3a4_veith       | TDC         | classification | 12,328    | 9,861  | 2,467 | predefined         | raw          | auprc          | Ensemble (OOF Stacking (RidgeCV, 5-fold))       | 0.905      | AUPRC              | 0.905       | 2         | 0.923          | MaxQsaring           |
| tdc_dili               | TDC         | classification | 475       | 379    | 96    | predefined         | raw          | roc_auc        | Ensemble (OOF Stacking (RidgeCV, 5-fold))       | 0.933      | ROC_-AUC           | 0.933       | 4         | 0.945          | Meta-model (NIST)    |
| tdc_herg               | TDC         | classification | 655       | 523    | 132   | predefined         | raw          | roc_auc        | Chemprop v2 (AttentiveFP, ensemble=3)           | 0.857      | ROC_-AUC           | 0.857       | 3         | 0.880          | MaxQsaring           |
| tdc_herg_karim         | TDC         | classification | 13,445    | 10,755 | 2,690 | scaffold           | raw          | roc_auc        | Ensemble (Weighted average (inverse OOF error)) | 0.907      |                    |             |           |                |                      |
| tdc_hia_hou            | TDC         | classification | 578       | 461    | 117   | predefined         | raw          | roc_auc        | XGBoost (ADMETboost features)                   | 0.994      | ROC_-AUC           | 0.994       | 2         | 0.994          | MiniMol (GINE)       |
| tdc_pampa_ncats        | TDC         | classification | 2,034     | 1,626  | 408   | scaffold           | raw          | roc_auc        | Uni-Mol V1                                      | 0.763      |                    |             |           |                |                      |
| tdc_pgp_broccatelli    | TDC         | classification | 1,218     | 973    | 245   | predefined         | raw          | roc_auc        | Uni-Mol V2 (84m)                                | 0.933      | ROC_-AUC           | 0.933       | 4         | 0.994          | MiniMol (GINE)       |
| tdc_skin_reaction      | TDC         | classification | 404       | 289    | 115   | scaffold           | raw          | roc_auc        | XGBoost (ADMETboost features)                   | 0.667      | ROC_-AUC           | 0.667       | >10       | 0.741          | FATE-Tox (MTL)       |
| tdc_tox21              | TDC         | classification | 7,258     | 5,797  | 1,461 | scaffold           | raw          | roc_auc        | Chemprop v2 (AttentiveFP, ensemble=3)           | 0.597      | ROC_-AUC           | 0.597       | >10       | 0.867          | PrismNet             |
| tdc_toxcast            | TDC         | classification | 1,731     | 1,357  | 374   | scaffold           | raw          | roc_auc        | HistGradient Boosting                           | 0.716      | ROC_-AUC           | 0.716       | 1         | 0.714          | PrismNet             |
| chemml_cep_homo        | ChemML      | regression     | 500       | 400    | 100   | target_-           | raw          | rmse           | Ensemble (OOF Stacking (RidgeCV, 5-fold))       | 0.090      |                    |             |           |                |                      |
| chemml_organic_density | ChemML      | regression     | 500       | 400    | 100   | quartiles target_- | log10        | rmse           | TabPFNRegressor                                 | 0.005      |                    |             |           |                |                      |
| esol_delaney           | MoleculeNet | regression     | 1,128     | 874    | 254   | quartiles scaffold | raw          | rmse           | Ensemble (Weighted average (inverse OOF error)) | 0.622      | RMSE               | 0.622       | 7         | 0.558          | GCN                  |
| freesolv_sampler       | MoleculeNet | regression     | 642       | 513    | 129   | random             | raw          | rmse           | TabPFNRegressor                                 | 0.933      |                    |             |           |                |                      |
| lipophilicity          | MoleculeNet | regression     | 4,200     | 3,357  | 843   | scaffold           | raw          | rmse           | Ensemble (OOF Stacking (RidgeCV, 5-fold))       | 0.507      | RMSE               | 0.507       | 1         | 0.549          | GCN                  |

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| Dataset                          | Suite   | Task       | Molecules | Train | Test | Split                  | Target scale | Ranking metric | Best model                                      | Best value | Leaderboard metric | QSArena (lb metric) | Est. rank | Best published | Best published model           |
|----------------------------------|---------|------------|-----------|-------|------|------------------------|--------------|----------------|-------------------------------------------------|------------|--------------------|---------------------|-----------|----------------|--------------------------------|
| poduam_pod_nc_std                | PODUAM  | regression | 1,842     | 1,473 | 369  | target_- raw           | raw          | rmse           | Ensemble (OOFS<br>Stacking (RidgeCV, 5-fold))   | 0.694      | RMSE               | 0.694               | 2         | 0.550          | PODUAM BNN (PODnc)             |
| poduam_pod_rd_std                | PODUAM  | regression | 2,355     | 1,884 | 471  | quartiles target_- raw | raw          | rmse           | Ensemble (OOFS<br>Stacking (RidgeCV, 5-fold))   | 0.556      | RMSE               | 0.556               | 2         | 0.410          | PODUAM BNN (PODr)              |
| polaris_adme_fang_hppb_1         | Polaris | regression | 1,808     | 1,446 | 362  | quartiles predefined   | raw          | rmse           | Ensemble (OOFS<br>Stacking (RidgeCV, 5-fold))   | 0.422      | MSE                | 0.178               | 4         | 0.143          | LateFusion-k5                  |
| polaris_adme_fang_perm_1         | Polaris | regression | 2,642     | 2,113 | 529  | predefined             | raw          | rmse           | Ensemble (OOFS<br>Stacking (RidgeCV, 5-fold))   | 0.379      | MSE                | 0.144               | 4         | 0.113          | 1B_MPNN_MolGPS-ens_Large Mix   |
| polaris_adme_fang_rclint_1       | Polaris | regression | 3,054     | 2,443 | 611  | predefined             | raw          | rmse           | Ensemble (OOFS<br>Stacking (RidgeCV, 5-fold))   | 0.484      | MSE                | 0.234               | 5         | 0.204          | LateFusion-k5                  |
| polaris_adme_fang_rppb_1         | Polaris | regression | 885       | 708   | 177  | predefined             | raw          | rmse           | Ensemble (OOFS<br>Stacking (RidgeCV, 5-fold))   | 0.478      | MSE                | 0.228               | 1         | 0.230          | 1B_MPNN_LargeMix-and-Phenomics |
| polaris_adme_fang_solu_1         | Polaris | regression | 2,173     | 1,738 | 435  | predefined             | raw          | rmse           | Uni-Mol V2 (84m)                                | 0.551      | MSE                | 0.304               | 7         | 0.222          | 1B_MPNN_MolGPS-ens_Large Mix   |
| tdc_caco2_wang                   | TDC     | regression | 910       | 728   | 182  | predefined             | raw          | rmse           | Ensemble (OOFS<br>Stacking (RidgeCV, 5-fold))   | 0.338      | MAE                | 0.269               | 3         | 0.256          | CaliciBoost                    |
| tdc_clearance_hepatocyte_az      | TDC     | regression | 1,213     | 970   | 243  | predefined             | raw          | rmse           | MapLight + GNN (CatBoost, Strict Parity)        | 44.113     | SPEARMAN           | 0.516               | 3         | 0.633          | CFA                            |
| tdc_clearance_microsome_az       | TDC     | regression | 1,102     | 881   | 221  | predefined             | raw          | rmse           | Uni-Mol V2 (84m)                                | 33.507     | SPEARMAN           | 0.678               | 1         | 0.652          | MapLight + GNN                 |
| tdc_half_life_obach              | TDC     | regression | 667       | 532   | 135  | predefined             | raw          | rmse           | Uni-Mol V1                                      | 19.128     | SPEARMAN           | 0.597               | 2         | 0.649          | CFA                            |
| tdc_hydrationfreeenergy_freesolv | TDC     | regression | 642       | 490   | 152  | scaffold               | raw          | rmse           | Ensemble (Weighted average (inverse OOF error)) | 1.075      | RMSE               | 1.075               | 7         | 0.654          | PrismNet                       |

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| Dataset                               | Suite | Task       | Molecules | Train | Test  | Split      | Target<br>scale | Ranking<br>met-<br>ric | Best model                                      | Best value | Leaderboard<br>metric | QSARena<br>(lb<br>metric) | Est.<br>rank | Best published | Best<br>published<br>model              |
|---------------------------------------|-------|------------|-----------|-------|-------|------------|-----------------|------------------------|-------------------------------------------------|------------|-----------------------|---------------------------|--------------|----------------|-----------------------------------------|
| tdc_ld50_<br>zhu                      | TDC   | regression | 7,385     | 5,907 | 1,478 | predefined | raw             | rmse                   | TabPFNRegressor                                 | 0.806      | MAE                   | 0.552                     | 2            | 0.292          | Base<br>Boosting<br>KyQVZ6b2<br>MiniMol |
| tdc_<br>lipophilicity_<br>astrazeneca | TDC   | regression | 4,200     | 3,360 | 840   | predefined | raw             | rmse                   | Ensemble (OOF<br>Stacking<br>(RidgeCV, 5-fold)) | 0.551      | MAE                   | 0.412                     | 3            | 0.406          | MiniMol                                 |
| tdc_ppbr_az                           | TDC   | regression | 2,790     | 2,231 | 559   | predefined | raw             | rmse                   | Ensemble (OOF<br>Stacking<br>(RidgeCV, 5-fold)) | 10.768     | MAE                   | 7.118                     | 2            | 0.679          | Gradient<br>Boost                       |
| tdc_<br>solubility_<br>aqsoldb        | TDC   | regression | 9,980     | 7,985 | 1,995 | predefined | raw             | rmse                   | Ensemble (OOF<br>Stacking<br>(RidgeCV, 5-fold)) | 0.955      | MAE                   | 0.676                     | 2            | 0.557          | MiniMol                                 |
| tdc_vdss_<br>lombardo                 | TDC   | regression | 1,130     | 904   | 226   | predefined | raw             | rmse                   | MapLight + GNN<br>(CatBoost, Strict<br>Parity)  | 4.670      | SPEARMAN              | 0.680                     | 5            | 0.942          | MapLight +<br>GNN                       |

Table S7: Per-dataset leaderboard comparison, sorted by estimated rank. “References (n)” is the number of published values available for that dataset and metric; ranks from sparse reference sets are correspondingly uncertain. The **tdc\_ppbr\_az** top-1 reference (MAE 0.679) is inconsistent in scale with the rest of that dataset’s references (top-10 cutoff 7.914) and its top-1 gap should be disregarded.

| Dataset                    | Metric    | QSARena best model (test-selected)        | QSARena value | Reference top-1 | Reference top-10 cutoff | References (n) | Est. rank | CV-selected model             | CV-selected value | CV-selected est. rank |
|----------------------------|-----------|-------------------------------------------|---------------|-----------------|-------------------------|----------------|-----------|-------------------------------|-------------------|-----------------------|
| lipophilicity              | RMSE      | Ensemble (OOF Stacking (RidgeCV, 5-fold)) | 0.507         | 0.549           | 0.610                   | 27             | 1         | TabPFNRegressor               | 0.595             | 8                     |
| polaris_adme_fang_rppb_1   | MSE       | Ensemble (OOF Stacking (RidgeCV, 5-fold)) | 0.228         | 0.230           | 0.634                   | 10             | 1         | XGBoost (ADMETboost features) | 0.257             | 3                     |
| tdc_bioavailability_ma     | ROC_- AUC | CatBoost                                  | 0.777         | 0.748           | 0.640                   | 8              | 1         | XGBoost (ADMETboost features) | 0.721             | 2                     |
| tdc_carcinogens_lagunin    | ROC_- AUC | Chemprop v2 (AttentiveFP, ensemble=3)     | 0.929         | 0.848           | 0.795                   | 20             | 1         | XGBoost (ADMETboost features) | 0.908             | 1                     |
| tdc_clearance_microsome_az | SPEARMA   | AutoMol V2 (84m)                          | 0.678         | 0.652           | 0.599                   | 17             | 1         | XGBoost (ADMETboost features) | 0.549             | >10                   |
| tdc_toxcast                | ROC_- AUC | HistGradientBoosting                      | 0.716         | 0.714           | 0.714                   | 2              | 1         | XGBoost (ADMETboost features) | 0.704             | 3                     |
| poduam_pod_nc_std          | RMSE      | Ensemble (OOF Stacking (RidgeCV, 5-fold)) | 0.694         | 0.550           | 0.730                   | 2              | 2         | XGBoost (ADMETboost features) | 0.729             | 2                     |
| poduam_pod_rd_std          | RMSE      | Ensemble (OOF Stacking (RidgeCV, 5-fold)) | 0.556         | 0.410           | 0.630                   | 2              | 2         | XGBoost (ADMETboost features) | 0.567             | 2                     |
| tdc_ames                   | ROC_- AUC | Ensemble (OOF Stacking (RidgeCV, 5-fold)) | 0.883         | 0.912           | 0.834                   | 7              | 2         | Extra trees                   | 0.871             | 2                     |
| tdc_bbb_martins            | ROC_- AUC | Random forest                             | 0.925         | 0.941           | 0.903                   | 7              | 2         | XGBoost (ADMETboost features) | 0.916             | 2                     |
| tdc_cyp2d6_veith           | AUPRC     | Ensemble (OOF Stacking (RidgeCV, 5-fold)) | 0.763         | 0.811           | 0.464                   | 6              | 2         | AdaBoost                      | 0.615             | 6                     |
| tdc_cyp3a4_veith           | AUPRC     | Ensemble (OOF Stacking (RidgeCV, 5-fold)) | 0.905         | 0.923           | 0.750                   | 6              | 2         | AdaBoost                      | 0.798             | 6                     |
| tdc_half_life_obach        | SPEARMA   | AutoMol V1                                | 0.597         | 0.649           | 0.485                   | 16             | 2         | XGBoost (ADMETboost features) | 0.473             | >10                   |

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| Dataset                            | Metric   | QSARena best model<br>(test-selected)      | QSARena value | Reference top-1 | Reference<br>top-10<br>cutoff | References (n) | Est.<br>rank | CV-selected<br>model          | CV-selected<br>value | CV-<br>selected<br>est. rank |
|------------------------------------|----------|--------------------------------------------|---------------|-----------------|-------------------------------|----------------|--------------|-------------------------------|----------------------|------------------------------|
| tdc_hia_hou                        | ROC_AUC  | XGBoost (ADMETboost features)              | 0.994         | 0.994           | 0.976                         | 8              | 2            | XGBoost (ADMETboost features) | 0.994                | 2                            |
| tdc_ld50_zhu                       | MAE      | TabPFNRegressor                            | 0.552         | 0.292           | 0.605                         | 16             | 2            | XGBoost                       | 0.588                | 8                            |
| tdc_ppbr_az                        | MAE      | Ensemble (OOF Stacking (RidgeCV, 5-fold))  | 7.118         | 0.679           | 7.914                         | 16             | 2            | TabPFNRegressor               | 8.537                | >10                          |
| tdc_solubility_aqsolddb            | MAE      | Ensemble (OOF Stacking (RidgeCV, 5-fold))  | 0.676         | 0.557           | 0.776                         | 17             | 2            | XGBoost (ADMETboost features) | 0.719                | 2                            |
| tdc_caco2_wang                     | MAE      | Ensemble (OOF Stacking (RidgeCV, 5-fold))  | 0.269         | 0.256           | 0.288                         | 20             | 3            | XGBoost (ADMETboost features) | 0.293                | >10                          |
| tdc_clearance_hepatocyte_az        | SPEARMAN | MANLight + GNN (Cat Boost, Strict Parity)  | 0.516         | 0.633           | 0.440                         | 16             | 3            | XGBoost (ADMETboost features) | 0.369                | >10                          |
| tdc_clintox                        | ROC_AUC  | CFA (Combinatorial Fusion)                 | 0.974         | 0.996           | 0.889                         | 12             | 3            | AdaBoost                      | 0.886                | >10                          |
| tdc_cyp2c9_substrate_carbonmangels | AUPRC    | Chemprop v2 (AttentiveFP, ensemble=3)      | 0.438         | 0.450           | 0.360                         | 6              | 3            | SVC                           | 0.301                | 7                            |
| tdc_cyp2c9_veith                   | AUPRC    | Ensemble (OOF Stacking (RidgeCV, 5-fold))  | 0.827         | 0.877           | 0.770                         | 6              | 3            | AdaBoost                      | 0.705                | 7                            |
| tdc_herg                           | ROC_AUC  | Chemprop v2 (AttentiveFP, ensemble=3)      | 0.857         | 0.880           | 0.806                         | 7              | 3            | XGBoost (ADMETboost features) | 0.792                | 8                            |
| tdc_lipophilicity_astrazeneca      | MAE      | Ensemble (OOF Stacking (RidgeCV, 5-fold))  | 0.412         | 0.406           | 0.515                         | 17             | 3            | TabPFNRegressor               | 0.439                | 4                            |
| polaris_adme_fang_hppb_1           | MSE      | Ensemble (OOF Stacking (RidgeCV, 5-fold))  | 0.178         | 0.143           | 0.303                         | 10             | 4            | XGBoost (ADMETboost features) | 0.214                | 5                            |
| polaris_adme_fang_perm_1           | MSE      | Ensemble (OOF Stacking (RidgeCV, 5-fold))  | 0.144         | 0.113           | 0.239                         | 10             | 4            | XGBoost (ADMETboost features) | 0.194                | 8                            |
| tdc_cyp2d6_substrate_carbonmangels | AUPRC    | Chemprop v2 (D-MPNN + RDKit2D, ensemble=3) | 0.693         | 0.766           | 0.570                         | 7              | 4            | ChemML MLP (PyTorch)          | 0.512                | 8                            |
| tdc_dili                           | ROC_AUC  | Ensemble (OOF Stacking (RidgeCV, 5-fold))  | 0.933         | 0.945           | 0.852                         | 6              | 4            | XGBoost (ADMETboost features) | 0.923                | 4                            |
| tdc_pgp_broccatelli                | ROC_AUC  | Uni-Mol V2 (84m)                           | 0.933         | 0.994           | 0.911                         | 8              | 4            | Random forest                 | 0.904                | 9                            |
| polaris_adme_fang_rcint_1          | MSE      | Ensemble (OOF Stacking (RidgeCV, 5-fold))  | 0.234         | 0.204           | 0.337                         | 10             | 5            | XGBoost                       | 0.293                | 6                            |

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| Dataset                                    | Metric      | QSARena best model<br>(test-selected)                 | QSARena value | Reference top-1 | Reference<br>top-10<br>cutoff | References (n) | Est.<br>rank | CV-selected<br>model                | CV-selected<br>value | CV-<br>selected<br>est. rank |
|--------------------------------------------|-------------|-------------------------------------------------------|---------------|-----------------|-------------------------------|----------------|--------------|-------------------------------------|----------------------|------------------------------|
| tdc_vdss_<br>lombardo                      | SPEARMAN    | MapLight + GNN (Cat<br>Boost, Strict Parity)          | 0.680         | 0.942           | 0.582                         | 16             | 5            | XGBoost<br>(ADMETboost<br>features) | 0.465                | >10                          |
| tdc_cyp3a4_<br>substrate_<br>carbonmangels | ROC_<br>AUC | Ensemble (OOF Stacking<br>(RidgeCV, 5-fold))          | 0.665         | 0.692           | 0.651                         | 7              | 6            | XGBoost<br>(ADMETboost<br>features) | 0.660                | 7                            |
| esol_delaney                               | RMSE        | Ensemble (Weighted<br>average (inverse OOF<br>error)) | 0.622         | 0.558           | 0.743                         | 30             | 7            | TabPFNRegressor                     | 0.673                | 9                            |
| polaris_adme_<br>fang_solu_1               | MSE         | Uni-Mol V2 (84m)                                      | 0.304         | 0.222           | 0.323                         | 10             | 7            | XGBoost<br>(ADMETboost<br>features) | 0.353                | >10                          |
| tdc_<br>hydrationfreeenergy_<br>freesolv   | RMSE        | Ensemble (Weighted<br>average (inverse OOF<br>error)) | 1.075         | 0.654           | 1.211                         | 12             | 7            | XGBoost<br>(ADMETboost<br>features) | 1.138                | 9                            |
| tdc_skin_reaction                          | ROC_<br>AUC | XGBoost (ADMETboost<br>features)                      | 0.667         | 0.741           | 0.677                         | 21             | >10          | XGBoost<br>(ADMETboost<br>features) | 0.667                | >10                          |
| tdc_tox21                                  | ROC_<br>AUC | Chemprop v2<br>(AttentiveFP, ensemble=3)              | 0.597         | 0.867           | 0.840                         | 12             | >10          | Extra trees                         | 0.549                | >10                          |

Table S8: Applicability domain and calibration on the 22 official TDC ADMET Benchmark Group splits, for a fixed reference model (random forest, 300 trees, on Morgan radius-2 2048-bit fingerprints plus RDKit 2D descriptors) fitted to 80% of each train\_val partition, with the remaining 20% used only for conformal calibration. Coverage is the fraction of test compounds inside the domain. The error ratio is the mean test error outside the domain divided by that inside it (MAE for regression, 0/1 error for classification); values above 1 mean the domain flag separates worse predictions. “Std.” is the Roy standardization approach on RDKit descriptors; “kNN” is the 5-nearest-neighbour Tanimoto distance thresholded at the 95th percentile of training distances; “Reliability” is the conformal-interval-width flag (regression) or the posterior-probability confidence flag with  $|p_1 - p_0| \geq 0.5$  (classification). Conformal coverage is the empirical test coverage of 90% split-conformal intervals or prediction sets. “—” marks a ratio not reported because fewer than five test compounds fall on one side of the domain boundary. Results describe the reference model, not the benchmark’s selected model; single seed.

| Dataset                        | Task | n test | Std. coverage | Std. ratio | kNN coverage | kNN ratio | Reliability ratio | Conformal coverage | ECE  |
|--------------------------------|------|--------|---------------|------------|--------------|-----------|-------------------|--------------------|------|
| caco2_wang                     | reg. | 182    | 0.99          | —          | 0.74         | 0.75      | 1.38              | 0.95               | —    |
| clearance_hepatocyte_az        | reg. | 243    | 0.94          | 1.09       | 0.88         | 1.26      | 1.75              | 0.94               | —    |
| clearance_microsome_az         | reg. | 221    | 0.99          | —          | 0.97         | 1.32      | 2.08              | 0.93               | —    |
| half_life_obach                | reg. | 135    | 0.95          | 1.36       | 0.94         | 0.87      | 3.60              | 0.94               | —    |
| ld50_zhu                       | reg. | 1478   | 0.95          | 1.16       | 0.82         | 1.24      | 2.11              | 0.82               | —    |
| lipophilicity_astazeneca       | reg. | 840    | 0.99          | 2.33       | 0.94         | 1.55      | 1.47              | 0.88               | —    |
| ppbr_az                        | reg. | 559    | 0.99          | 1.35       | 0.91         | 1.05      | 3.01              | 0.93               | —    |
| solubility_aqsolddb            | reg. | 1995   | 0.95          | 1.45       | 0.86         | 1.15      | 1.59              | 0.89               | —    |
| vdss_lombardo                  | reg. | 226    | 0.94          | 0.49       | 0.86         | 1.22      | 8.26              | 0.95               | —    |
| ames                           | cls. | 1457   | 0.96          | 0.91       | 0.84         | 1.71      | 2.67              | 0.89               | 0.07 |
| bbb_martins                    | cls. | 406    | 0.96          | 0.00       | 0.78         | 1.19      | 9.30              | 0.91               | 0.06 |
| bioavailability_ma             | cls. | 128    | 0.98          | —          | 0.96         | 0.72      | 3.72              | 0.91               | 0.08 |
| cyp2c9_substrate_carbonmangels | cls. | 135    | 0.93          | 0.68       | 0.95         | 0.00      | 2.15              | 0.81               | 0.11 |
| cyp2c9_veith                   | cls. | 2419   | 0.96          | 0.53       | 0.88         | 0.62      | 5.21              | 0.91               | 0.05 |
| cyp2d6_substrate_carbonmangels | cls. | 135    | 0.96          | 0.65       | 0.98         | —         | 4.82              | 0.89               | 0.06 |
| cyp2d6_veith                   | cls. | 2626   | 0.96          | 0.90       | 0.84         | 0.89      | 5.29              | 0.91               | 0.06 |
| cyp3a4_substrate_carbonmangels | cls. | 135    | 0.95          | 2.08       | 0.94         | 1.80      | 1.47              | 0.93               | 0.06 |
| cyp3a4_veith                   | cls. | 2467   | 0.98          | 0.91       | 0.94         | 0.93      | 4.88              | 0.89               | 0.07 |
| dili                           | cls. | 96     | 0.95          | 3.64       | 0.95         | 0.00      | 2.16              | 0.95               | 0.17 |
| herg                           | cls. | 132    | 0.99          | —          | 0.92         | 1.10      | 4.71              | 0.86               | 0.10 |
| hia_hou                        | cls. | 117    | 0.97          | —          | 0.82         | 4.57      | 6.13              | 0.91               | 0.09 |
| pgp_broccatelli                | cls. | 245    | 0.94          | 0.75       | 0.84         | 1.21      | 5.04              | 0.88               | 0.06 |

Table S9: Dataset meta-feature catalog computed on each benchmark train/test partition (verified against the recorded split hashes). Diversity and similarity use Morgan radius-2, 2,048-bit fingerprints; SNN is the maximum Tanimoto similarity of a test molecule to any training molecule, and OOD fraction is the share of test molecules with SNN below 0.40.

| Dataset                 | Task           | Split            | Metric | n train | n test | Positive fraction | Imbalance ratio | Target SD | Target skew | Scaffolds | Scaffolds / molecule | Singleton scaffold fraction | Internal diversity | Mean SNN | OOD fraction (SNN < 0.40) |
|-------------------------|----------------|------------------|--------|---------|--------|-------------------|-----------------|-----------|-------------|-----------|----------------------|-----------------------------|--------------------|----------|---------------------------|
| tdc_ames                | classification | patidefinedest_- | auc    | 5,821   | 1,457  | 0.533             | 1.142           |           |             | 2,387     | 0.410                | 0.809                       | 0.916              | 0.517    | 0.251                     |
|                         |                |                  | roc_-  |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_bbb_martins         | classification | patidefinedest_- | auc    | 1,624   | 406    | 0.753             | 3.050           |           |             | 829       | 0.510                | 0.744                       | 0.890              | 0.418    | 0.520                     |
|                         |                |                  | roc_-  |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_bioavailability_ma  | classification | patidefinedest_- | auc    | 512     | 128    | 0.771             | 3.376           |           |             | 363       | 0.709                | 0.837                       | 0.899              | 0.435    | 0.492                     |
|                         |                |                  | roc_-  |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_carcinogens_lagunin | classification | scaffolddest_-   | auc    | 223     | 57     | 0.184             | 4.439           |           |             | 191       | 0.857                | 0.864                       | 0.914              | 0.346    | 0.684                     |
|                         |                |                  | roc_-  |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_clintox             | classification | scaffolddest_-   | auc    | 1,180   | 298    | 0.083             | 11.041          |           |             | 894       | 0.758                | 0.813                       | 0.915              | 0.425    | 0.517                     |
|                         |                |                  | roc_-  |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_cyp1a2_veith        | classification | scaffolddest_-   | auc    | 10,061  | 2,518  | 0.445             | 1.249           |           |             | 7,622     | 0.758                | 0.811                       | 0.887              | 0.561    | 0.199                     |
|                         |                |                  | roc_-  |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_cyp2c19_veith       | classification | scaffolddest_-   | auc    | 10,131  | 2,534  | 0.496             | 1.018           |           |             | 7,615     | 0.752                | 0.807                       | 0.887              | 0.563    | 0.204                     |
|                         |                |                  | roc_-  |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
|                         |                |                  | auc    |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |

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| Dataset        | Task           | Split         | Metric | n train | n test | Positive fraction | Imbalance ratio | Target SD | Target skew | Scaffolds | Scaffolds / molecule | Singleton scaffold fraction | Internal diversity | Mean SNN | OOD fraction (SNN < 0.40) |
|----------------|----------------|---------------|--------|---------|--------|-------------------|-----------------|-----------|-------------|-----------|----------------------|-----------------------------|--------------------|----------|---------------------------|
| tdc_cyp2c9_-   | classification | patidefined_- | auprc  | 534     | 135    | 0.193             | 4.184           |           |             | 366       | 0.685                | 0.831                       | 0.899              | 0.408    | 0.504                     |
| substrate_-    |                |               |        |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| carbonmangels  |                |               |        |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_cyp2c9_-   | classification | patidefined_- | auprc  | 9,673   | 2,419  | 0.339             | 1.954           |           |             | 5,727     | 0.592                | 0.784                       | 0.885              | 0.509    | 0.338                     |
| veith          |                |               |        |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_cyp2d6_-   | classification | patidefined_- | auprc  | 532     | 135    | 0.278             | 2.595           |           |             | 373       | 0.701                | 0.845                       | 0.898              | 0.426    | 0.526                     |
| substrate_-    |                |               |        |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| carbonmangels  |                |               |        |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_cyp2d6_-   | classification | patidefined_- | auprc  | 10,504  | 2,626  | 0.197             | 4.072           |           |             | 6,643     | 0.632                | 0.779                       | 0.884              | 0.487    | 0.364                     |
| veith          |                |               |        |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_cyp3a4_-   | classification | patidefined_- | auprc  | 535     | 135    | 0.516             | 1.066           |           |             | 375       | 0.701                | 0.832                       | 0.899              | 0.436    | 0.474                     |
| substrate_-    |                |               |        |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| carbonmangels  |                |               |        |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_cyp3a4_-   | classification | patidefined_- | auprc  | 9,861   | 2,467  | 0.408             | 1.448           |           |             | 6,088     | 0.617                | 0.800                       | 0.889              | 0.534    | 0.260                     |
| veith          |                |               |        |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_dili       | classification | patidefined_- | roc_-  | 379     | 96     | 0.491             | 1.038           |           |             | 265       | 0.699                | 0.879                       | 0.907              | 0.332    | 0.781                     |
| tdc_herg       | classification | patidefined_- | auc    | 523     | 132    | 0.677             | 2.095           |           |             | 341       | 0.652                | 0.792                       | 0.895              | 0.484    | 0.379                     |
| tdc_herg_karim | classification | patidefined_- | auc    | 10,755  | 2,690  | 0.492             | 1.031           |           |             | 5,884     | 0.547                | 0.646                       | 0.878              | 0.672    | 0.084                     |
|                |                |               | auc    |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |

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| Dataset                | Task           | Split    | Metric      | n train | n test | Positive fraction | Imbalance ratio | Target SD | Target skew | Scaffolds | Scaffolds / molecule | Singleton scaffold fraction | Internal diversity | Mean SNN | OOD fraction (SNN < 0.40) |
|------------------------|----------------|----------|-------------|---------|--------|-------------------|-----------------|-----------|-------------|-----------|----------------------|-----------------------------|--------------------|----------|---------------------------|
| tdc_hia_hou            | classification | patidef  | roc_test_-  | 461     | 117    | 0.889             | 8.039           |           |             | 317       | 0.688                | 0.868                       | 0.894              | 0.371    | 0.598                     |
|                        |                |          | auc         |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_pampa_ncats        | classification | scaffold | roc_test_-  | 1,626   | 408    | 0.846             | 5.504           |           |             | 1,355     | 0.833                | 0.862                       | 0.871              | 0.642    | 0.064                     |
|                        |                |          | auc         |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_pgp_broccatelli    | classification | patidef  | roc_test_-  | 973     | 245    | 0.541             | 1.177           |           |             | 546       | 0.561                | 0.764                       | 0.874              | 0.490    | 0.400                     |
|                        |                |          | auc         |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_skin_reaction      | classification | scaffold | roc_test_-  | 289     | 115    | 0.661             | 1.949           |           |             | 247       | 0.855                | 0.895                       | 0.918              | 0.312    | 0.904                     |
|                        |                |          | auc         |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_tox21              | classification | scaffold | roc_test_-  | 5,797   | 1,461  | 0.049             | 19.412          |           |             | 3,853     | 0.665                | 0.872                       | 0.925              | 0.467    | 0.305                     |
|                        |                |          | auc         |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| tdc_toxcast            | classification | scaffold | roc_test_-  | 1,357   | 374    | 0.262             | 2.812           |           |             | 1,002     | 0.738                | 0.889                       | 0.926              | 0.439    | 0.422                     |
|                        |                |          | auc         |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| chemml_cep_homo        | regression     | target   | rmse_test_- | 400     | 100    |                   |                 | 0.687     | 0.029       | 400       | 1                    | 1                           | 0.844              | 0.492    | 0.220                     |
|                        |                |          | quartiles   |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| chemml_organic_density | regression     | target   | rmse_test_- | 400     | 100    |                   |                 | 0.031     | -0.093      | 399       | 0.998                | 0.997                       | 0.841              | 0.463    | 0.250                     |
|                        |                |          | quartiles   |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| esol_delaney           | regression     | scaffold | rmse_test_- | 874     | 254    |                   |                 | 2.247     | -0.435      | 583       | 0.667                | 0.873                       | 0.930              | 0.437    | 0.417                     |
|                        |                |          | rmse        |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |
| freesolv_sampler       | regression     | random   | rmse_test_- | 513     | 129    |                   |                 | 3.677     | -0.753      | 308       | 0.600                | 0.932                       | 0.923              | 0.570    | 0.163                     |
|                        |                |          | rmse        |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |

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| Dataset                    | Task       | Split     | Metric    | n train | n test | Positive fraction | Imbalance ratio | Target SD | Target skew | Scaffolds | Scaffolds / molecule | Singleton scaffold fraction | Internal diversity | Mean SNN | OOD fraction (SNN < 0.40) |
|----------------------------|------------|-----------|-----------|---------|--------|-------------------|-----------------|-----------|-------------|-----------|----------------------|-----------------------------|--------------------|----------|---------------------------|
| lipophilicity              | regression | scaffold  | test_rmse | 3,357   | 843    |                   |                 | 1.190     | -0.557      | 2,355     | 0.702                | 0.791                       | 0.882              | 0.584    | 0.152                     |
| poduam_nc_std              | regression | scaffold  | test_rmse | 1,473   | 369    |                   |                 | 1.109     | -1.529      | 945       | 0.642                | 0.899                       | 0.922              | 0.521    | 0.322                     |
| poduam_nc_std              | regression | quartiles | test_rmse | 1,884   | 471    |                   |                 | 0.865     | -1.743      | 1,237     | 0.657                | 0.905                       | 0.917              | 0.549    | 0.268                     |
| polaris_adme_fang_hppb_1   | regression | undefined | test_rmse | 1,446   | 362    |                   |                 | 0.719     | -0.337      | 1,003     | 0.694                | 0.827                       | 0.878              | 0.583    | 0.331                     |
| polaris_adme_fang_perm_1   | regression | undefined | test_rmse | 2,113   | 529    |                   |                 | 0.682     | 0.746       | 1,885     | 0.892                | 0.926                       | 0.875              | 0.443    | 0.465                     |
| polaris_adme_fang_rclint_1 | regression | undefined | test_rmse | 2,443   | 611    |                   |                 | 0.751     | -0.134      | 2,146     | 0.878                | 0.922                       | 0.876              | 0.444    | 0.435                     |
| polaris_adme_fang_rppb_1   | regression | undefined | test_rmse | 708     | 177    |                   |                 | 0.751     | -0.435      | 569       | 0.804                | 0.872                       | 0.877              | 0.511    | 0.429                     |
| polaris_adme_fang_solu_1   | regression | undefined | test_rmse | 1,738   | 435    |                   |                 | 0.668     | -1.657      | 1,531     | 0.881                | 0.925                       | 0.874              | 0.446    | 0.492                     |
| tdc_caco2_wang             | regression | undefined | test_rmse | 728     | 182    |                   |                 | 0.797     | -0.715      | 402       | 0.552                | 0.694                       | 0.890              | 0.459    | 0.473                     |
| tdc_clearance              | regression | undefined | test_rmse | 970     | 243    |                   |                 | 50.287    | 1.216       | 601       | 0.620                | 0.687                       | 0.877              | 0.425    | 0.580                     |
| hepatocyte_az              | regression | undefined | test_rmse | 881     | 221    |                   |                 | 45.128    | 1.548       | 587       | 0.666                | 0.818                       | 0.871              | 0.520    | 0.321                     |
| microsome_az               |            |           |           |         |        |                   |                 |           |             |           |                      |                             |                    |          |                           |

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| Dataset                          | Task       | Split     | Metric | n train | n test | Positive fraction | Imbalance ratio | Target SD | Target skew | Scaffolds | Scaffolds / molecule | Singleton scaffold fraction | Internal diversity | Mean SNN | OOD fraction (SNN < 0.40) |
|----------------------------------|------------|-----------|--------|---------|--------|-------------------|-----------------|-----------|-------------|-----------|----------------------|-----------------------------|--------------------|----------|---------------------------|
| tdc_half_life_obach              | regression | undefined | rmse   | 532     | 135    |                   |                 | 90.920    | 9.338       | 355       | 0.667                | 0.820                       | 0.899              | 0.413    | 0.533                     |
| tdc_hydrationfreeenergy_freesolv | regression | scaffold  | rmse   | 490     | 152    |                   |                 | 4.014     | -1.427      | 381       | 0.778                | 0.934                       | 0.927              | 0.384    | 0.625                     |
| tdc_ld50_zhu                     | regression | undefined | rmse   | 5,907   | 1,478  |                   |                 | 0.921     | 0.928       | 3,136     | 0.531                | 0.894                       | 0.917              | 0.448    | 0.438                     |
| tdc_lipophilicity                | regression | undefined | rmse   | 3,360   | 840    |                   |                 | 1.204     | -0.582      | 1,932     | 0.575                | 0.769                       | 0.879              | 0.575    | 0.219                     |
| astrazeneca_tdc_ppbr_az          | regression | undefined | rmse   | 2,231   | 559    |                   |                 | 18.785    | -1.962      | 946       | 0.424                | 0.498                       | 0.879              | 0.520    | 0.349                     |
| tdc_solubility_aqsolddb          | regression | undefined | rmse   | 7,985   | 1,995  |                   |                 | 2.369     | -0.587      | 4,265     | 0.534                | 0.900                       | 0.922              | 0.475    | 0.327                     |
| tdc_vdss_lombardo                | regression | undefined | rmse   | 904     | 226    |                   |                 | 24.829    | 24.663      | 613       | 0.678                | 0.822                       | 0.900              | 0.384    | 0.619                     |

Table S10: Exploratory family  $\times$  meta-feature associations: Spearman correlation between each meta-feature and the relative gap of the family’s best model to the per-dataset best, with a dataset-bootstrap 95% CI (10,000 resamples), a two-sided label-permutation p-value and a Benjamini–Hochberg q-value over the whole grid. Positive  $\rho$  means the family falls further behind as the meta-feature increases.

| Model family                    | Meta-feature           | Datasets | Spearman rho | 95% CI         | Permutation p | BH q  |
|---------------------------------|------------------------|----------|--------------|----------------|---------------|-------|
| Conventional ML                 | log10_n_train          | 44       | 0.034        | -0.29 to 0.35  | 0.818         | 0.873 |
| Conventional ML                 | internal_diversity     | 44       | -0.329       | -0.59 to -0.01 | 0.028         | 0.238 |
| Conventional ML                 | mean_snn               | 44       | 0.325        | 0.01 to 0.58   | 0.034         | 0.238 |
| Conventional ML                 | ood_fraction           | 44       | -0.326       | -0.59 to -0.01 | 0.035         | 0.238 |
| Conventional ML                 | scaffolds_per_molecule | 44       | 0.139        | -0.20 to 0.43  | 0.367         | 0.710 |
| Conventional ML                 | label_asymmetry        | 44       | 0.204        | -0.10 to 0.46  | 0.179         | 0.461 |
| Ensemble (stacking / averaging) | log10_n_train          | 44       | -0.467       | -0.70 to -0.19 | 0.001         | 0.058 |
| Ensemble (stacking / averaging) | internal_diversity     | 44       | 0.207        | -0.14 to 0.52  | 0.180         | 0.461 |
| Ensemble (stacking / averaging) | mean_snn               | 44       | -0.393       | -0.66 to -0.09 | 0.008         | 0.202 |
| Ensemble (stacking / averaging) | ood_fraction           | 44       | 0.288        | -0.03 to 0.59  | 0.062         | 0.297 |
| Ensemble (stacking / averaging) | scaffolds_per_molecule | 44       | 0.147        | -0.14 to 0.43  | 0.336         | 0.710 |
| Ensemble (stacking / averaging) | label_asymmetry        | 44       | 0.275        | -0.03 to 0.54  | 0.070         | 0.308 |
| CFA combinatorial fusion        | log10_n_train          | 44       | -0.220       | -0.47 to 0.07  | 0.156         | 0.461 |
| CFA combinatorial fusion        | internal_diversity     | 44       | -0.325       | -0.59 to -0.01 | 0.033         | 0.238 |
| CFA combinatorial fusion        | mean_snn               | 44       | 0.125        | -0.21 to 0.43  | 0.421         | 0.710 |
| CFA combinatorial fusion        | ood_fraction           | 44       | -0.081       | -0.39 to 0.24  | 0.602         | 0.743 |
| CFA combinatorial fusion        | scaffolds_per_molecule | 44       | 0.216        | -0.10 to 0.50  | 0.159         | 0.461 |
| CFA combinatorial fusion        | label_asymmetry        | 44       | 0.108        | -0.19 to 0.40  | 0.486         | 0.710 |
| Uni-Mol (3D pretrained)         | log10_n_train          | 44       | -0.108       | -0.41 to 0.20  | 0.488         | 0.710 |
| Uni-Mol (3D pretrained)         | internal_diversity     | 44       | 0.146        | -0.21 to 0.49  | 0.348         | 0.710 |
| Uni-Mol (3D pretrained)         | mean_snn               | 44       | 0.081        | -0.25 to 0.40  | 0.604         | 0.743 |
| Uni-Mol (3D pretrained)         | ood_fraction           | 44       | -0.118       | -0.46 to 0.22  | 0.442         | 0.710 |
| Uni-Mol (3D pretrained)         | scaffolds_per_molecule | 44       | 0.102        | -0.23 to 0.41  | 0.516         | 0.729 |
| Uni-Mol (3D pretrained)         | label_asymmetry        | 44       | 0.075        | -0.26 to 0.41  | 0.628         | 0.753 |
| TabPFN (tabular foundation)     | log10_n_train          | 41       | 0.149        | -0.24 to 0.50  | 0.346         | 0.710 |
| TabPFN (tabular foundation)     | internal_diversity     | 41       | -0.242       | -0.54 to 0.10  | 0.130         | 0.461 |
| TabPFN (tabular foundation)     | mean_snn               | 41       | 0.098        | -0.21 to 0.40  | 0.540         | 0.730 |
| TabPFN (tabular foundation)     | ood_fraction           | 41       | -0.004       | -0.34 to 0.33  | 0.983         | 0.983 |
| TabPFN (tabular foundation)     | scaffolds_per_molecule | 41       | 0.071        | -0.28 to 0.41  | 0.661         | 0.755 |
| TabPFN (tabular foundation)     | label_asymmetry        | 41       | -0.126       | -0.42 to 0.18  | 0.428         | 0.710 |
| Chemprop v2 GNN                 | log10_n_train          | 42       | -0.008       | -0.31 to 0.30  | 0.965         | 0.983 |
| Chemprop v2 GNN                 | internal_diversity     | 42       | -0.050       | -0.37 to 0.28  | 0.752         | 0.840 |
| Chemprop v2 GNN                 | mean_snn               | 42       | 0.131        | -0.19 to 0.43  | 0.405         | 0.710 |
| Chemprop v2 GNN                 | ood_fraction           | 42       | -0.119       | -0.44 to 0.21  | 0.454         | 0.710 |

continued on next page

| Model family                 | Meta-feature           | Datasets | Spearman rho | 95% CI        | Permutation p | BH q  |
|------------------------------|------------------------|----------|--------------|---------------|---------------|-------|
| Chemprop v2 GNN              | scaffolds_per_molecule | 42       | 0.141        | -0.16 to 0.42 | 0.370         | 0.710 |
| Chemprop v2 GNN              | label_asymmetry        | 42       | 0.325        | -0.03 to 0.62 | 0.031         | 0.238 |
| MapLight + GNN               | log10_n_train          | 44       | -0.205       | -0.47 to 0.10 | 0.182         | 0.461 |
| MapLight + GNN               | internal_diversity     | 44       | 0.091        | -0.23 to 0.40 | 0.559         | 0.730 |
| MapLight + GNN               | mean_snn               | 44       | -0.071       | -0.38 to 0.24 | 0.643         | 0.753 |
| MapLight + GNN               | ood_fraction           | 44       | -0.113       | -0.42 to 0.22 | 0.467         | 0.710 |
| MapLight + GNN               | scaffolds_per_molecule | 44       | 0.291        | 0.01 to 0.54  | 0.052         | 0.278 |
| MapLight + GNN               | label_asymmetry        | 44       | -0.090       | -0.39 to 0.25 | 0.563         | 0.730 |
| Deep tabular NN (ChemML MLP) | log10_n_train          | 44       | -0.315       | -0.57 to 0.00 | 0.042         | 0.250 |
| Deep tabular NN (ChemML MLP) | internal_diversity     | 44       | -0.251       | -0.51 to 0.04 | 0.100         | 0.401 |
| Deep tabular NN (ChemML MLP) | mean_snn               | 44       | -0.017       | -0.33 to 0.31 | 0.912         | 0.952 |
| Deep tabular NN (ChemML MLP) | ood_fraction           | 44       | 0.039        | -0.31 to 0.36 | 0.807         | 0.873 |
| Deep tabular NN (ChemML MLP) | scaffolds_per_molecule | 44       | 0.224        | -0.10 to 0.51 | 0.141         | 0.461 |
| Deep tabular NN (ChemML MLP) | label_asymmetry        | 44       | 0.181        | -0.14 to 0.47 | 0.242         | 0.581 |

Table S11: Family selector (v2), leave-one-dataset-out regret: the relative gap (%) of the recommended family to the per-dataset best. Rows are pre-specified variants (model | feature blocks: F0 size, similarity, label asymmetry and task; F1 chemistry; F2 label landscape; F3 training-set CV landmarks); “sbs” always picks the family with the best record on the other datasets. The nested row chooses the variant inside an inner leave-one-dataset-out loop and is the only unbiased estimate; single variants are exploratory.

| Selector (model   feature blocks) | Mean regret (%) | Median regret (%) | Pick within 5% of best (%) | SBS-to-oracle gap closed |
|-----------------------------------|-----------------|-------------------|----------------------------|--------------------------|
| sbs                               | 3.310           | 0.031             | 84.091                     | 0                        |
| knn F0                            | 3.760           | 0.159             | 79.545                     | -0.136                   |
| ridge F0                          | 2.982           | 0                 | 86.364                     | 0.099                    |
| rf F0                             | 4.236           | 0.082             | 77.273                     | -0.280                   |
| knn F0+F1                         | 3.629           | 0                 | 81.818                     | -0.096                   |
| ridge F0+F1                       | 3.403           | 0                 | 81.818                     | -0.028                   |
| rf F0+F1                          | 3.650           | 1.086             | 81.818                     | -0.102                   |
| knn F0+F2                         | 3.728           | 0.159             | 81.818                     | -0.126                   |
| ridge F0+F2                       | 3.323           | 0                 | 84.091                     | -0.004                   |
| rf F0+F2                          | 4.163           | 0.159             | 75                         | -0.257                   |
| knn F0+F3                         | 2.739           | 0                 | 88.636                     | 0.173                    |
| ridge F0+F3                       | 3.133           | 0                 | 84.091                     | 0.054                    |
| rf F0+F3                          | 3.216           | 0                 | 84.091                     | 0.028                    |
| knn F0-F3                         | 3.375           | 0                 | 81.818                     | -0.020                   |
| ridge F0-F3                       | 4.525           | 0.031             | 81.818                     | -0.367                   |
| rf F0-F3                          | 3.516           | 0.051             | 81.818                     | -0.062                   |
| tree_cls F0                       | 4.837           | 0                 | 79.545                     | -0.461                   |
| nested (headline)                 | 3.115           | 0                 | 86.364                     | 0.059                    |

## Supplementary Note S1: Benchmark runs, ensemble reconstruction and computing environment

This note gives the full provenance that main-text Section 2.12 summarizes, including the run directories, commit and commands, which the main text leaves out.

The benchmark reported here was executed on NSF ACCESS Jetstream2 [1, 2] under allocation CIS261142, on a `g3.large` instance providing one NVIDIA A100-SXM4-40GB GPU (39.5 GB device memory, 108 streaming multiprocessors, compute capability 8.0) and 32 CPU cores, running Linux with all timestamps in UTC. The device inventory, CPU count, parallelism settings and `fp32` precision mode are recorded in the run’s `run_timing.json` and `run_config.json` and are reproduced in the deposited artifacts. The Chemprop and TabPFN repair run used the same node; the out-of-fold ensemble rebuild ran on the RTX 4060 workstation (main-text Section 2.12), and the wall-clock and cost figures in this paper come from the original A100 run. Main-text Section 3.10 additionally reports the same benchmark executed on a consumer laptop GPU, as a direct test of whether the results depend on datacentre hardware. That run’s artifacts are in `benchmark_results/benchmark_name_date`, the default output-directory name it was launched with; the name is kept so that the paths and checksums recorded in its manifests stay valid.

All figures, tables and numerical claims in this paper are regenerated from committed benchmark artifacts by a single command, `python portable_colab_qsar_bundle/render_manuscript_assets.py`, which re-executes the analysis notebook and writes the figures, tables and a `manuscript_numbers.json` containing every headline number quoted in the text. A companion script, `verify_manuscript_numbers.py`, asserts that the manuscript text still matches those artifacts.

Every reported result is backed by a complete artifact trail. Each analysed dataset retains metrics, selected-feature records, selector coefficients, split-signature hashes, per-stage runtimes, CFA candidate tables and ensemble weights; the run recorded a configuration signature, fixed seeds (random seed 13, Chemprop seed 42), a full benchmark profile, `fp32` precision and a GPU inventory identifying the A100 device; and a SHA-256 manifest covers repository-level and per-dataset artifacts. The benchmark artifacts analysed here were committed in repository revision `bbfb188`. All figures, tables and quoted numbers in this paper regenerate from those artifacts with one command (main-text Section 2.11), which also writes a machine-readable record of every headline number, and a companion script asserts that the manuscript text still matches it.

The reported results combine three runs that share one data, split, feature and feature-selection configuration, which the runner verifies through stage signatures. (i) The base benchmark ran once on the Jetstream2 A100 node described in main-text Section 2.11 (`benchmark_results/autoqsar_benchmark_20260623_153839`). In that run, a text-encoding fault in the Chemprop harness discarded most Chemprop results, and TabPFN was disabled. (ii) A repair run on the same node (`benchmark_results/qsarena_benchmark_chemprop_fixed`) was seeded with the first run’s per-model results and trained only what was missing: the five Chemprop variants, which then produced valid results on 38 to 42 of the 44 datasets, TabPFN (valid on 31 of 44 before the API’s daily limit stopped it) and one interrupted dataset. No base model was retrained. (iii) Every ensemble was then rebuilt from out-of-fold predictions (`benchmark_results/qsarena_benchmark_oof_ensemble`, from which all reported results are computed). Each candidate member received an out-of-fold prediction for every training molecule on the training split’s five cross-validation folds. Uni-Mol supplied the predictions saved from its own internal five-fold training; every other member was refitted on each fold with its reported configuration, on a workstation with a consumer NVIDIA RTX 4060 laptop GPU. Base-model results are unchanged from the source runs, which the release checks automatically, with the documented exceptions below. (iv) The descriptor model of main-text Section 2.6 was trained into this run with the same configuration as every other model, after it performed well in an exploratory analysis. We flag this because adding a model after seeing results can flatter test-selected comparisons. (v) Four binary datasets are catalogued with a regression metric (CYP1A2, CYP2C19, hERG-Karim and PAMPA). On them, ten conventional and gradient-boosting models saved hard class labels instead of probabilities, so their test AUROC was computed on labels. Those models were retrained on those datasets with probability outputs, which raised their test AUROC by a median of 0.06 to 0.11 per dataset. TabPFN

was retrained on PAMPA with the locally installed TabPFN package (version 2.6 weights); on the other three datasets its label-based results are withdrawn, because the local model does not fit the workstation GPU at their size (about 10,000 training molecules and 1,000 selected features). Rerunning them through the Prior Labs API, which produced the other TabPFN results, was not feasible either: the API is metered by tokens (training rows times features times model fits), and the runner’s estimate for one run of these datasets with cross-validation is 111 to 127 million tokens each, more than twenty times the API’s daily allowance of 5 million tokens per key. TabPFN therefore has no result on these three datasets. (vi) Finally, the cross-validated metrics and out-of-fold predictions of every model that uses the selected features were recomputed with feature selection nested inside each fold (main-text Section 2.5), and every ensemble was rebuilt once over the corrected 31-model pool. TabPFN is not an ensemble member, because most of its full-fit predictions came from the Prior Labs API while its fold refits used the local package. On hERG and bioavailability, the conventional and ChemML models have no saved predictions, so they receive nested cross-validated scores but are not ensemble members there. Combinatorial fusion uses no cross-validated scores and was rebuilt once over the 31-model pool.

The rebuild replaced two earlier ensemble protocols. The first run selected and weighted ensemble members by held-out test metrics, which leaks test information into the ensemble row. A second attempt used in-sample training predictions instead. That removed the leak but rewarded memorisation: tree ensembles that fit the training set almost exactly took nearly all of the stacking weight, and regression ensemble wins fell from 7 to 1. Under the out-of-fold protocol with nested feature selection, ensembles won 22 datasets (14 regression, 8 classification), against 16 (7 and 9) under the leaky first protocol and 12 (6 and 6) when their out-of-fold predictions came from the outer feature selection. The feature-selection leak had also inflated the out-of-fold predictions of the members that use selected features, so the stacker over-weighted them; nesting the selection improved the best ensemble on 39 of the 44 datasets (median error reduction 4.9% for regression; median AUROC gain 0.7% for classification), while no base model’s test result changed. The rebuild also fixed two faults that would otherwise have biased the ensembles. Uni-Mol’s saved fold predictions are on a normalised target scale, and its scaler file was not among the transferred artifacts, so the scaler is now rebuilt from the training targets. And a few members extrapolated far outside the target range on single molecules, which motivated the clipping described in main-text Section 2.7. Two Chemprop variants still lack out-of-fold predictions on Tox21, where Chemprop cannot featurise one training molecule, and are excluded there.

Two datasets need a note on coverage. A 45th dataset, `tdc_herg_central`, was abandoned for compute budget after more than 24 hours in the base run without completing, and is not analysed. `polaris_adme_fang_hppb_1` was interrupted when the base run ended and was completed by the repair run, so all 44 analysed datasets are complete.

## Supplementary Note S2: Molecular representations, feature store and model configurations

Ten configurable feature families were implemented on RDKit, all returning fixed-width numeric matrices. Circular and path-based fingerprints were generated at 1024 bits by default. Morgan fingerprints [3] used radius 2; ECFP6 and FCFP6 were generated with the RDKit MorganGenerator at radius 3, FCFP6 using feature-based atom invariants; RDKit layered, atom-pair, topological-torsion and RDKit path fingerprints were each computed at 1024 bits, the path fingerprint spanning path lengths 1–7. MACCS keys [4] were generated at their native 167-bit width, and the full RDKit 2D descriptor set was computed from `Descriptors._descList`.

The “MapLight classic” family reproduces the feature construction of the MapLight TDC submission [5]: a hashed Morgan count fingerprint (radius 2, 1024 bits), an Avalon count fingerprint (1024 bits) [6], the extended reduced graph (ErG) fingerprint [7], and a curated panel of approximately 200 RDKit descriptors spanning connectivity and shape indices (BalabanJ, BertzCT, Chi and Kappa indices), VSA-family descriptors, physicochemical properties (ExactMolWt, MolLogP, TPSA, FractionCSP3), hydrogen-bonding and ring counts, functional-group fragment counts, and the QED drug-likeness score. In the feature-selection analyses below, this composite appears as its three constituent families (`avalon`, `erg` and the `maplight` descriptor panel), and we report both the components and their sum.

After assembly, every feature matrix passed a common finalization step: infinities replaced with missing values, all columns coerced to numeric, values exceeding the float32 range masked, remaining gaps imputed with the per-column median (falling back to zero), and the matrix cast to float32.

To avoid recomputing expensive descriptors across datasets and runs, features were cached in a persistent, content-addressed store writing Parquet shards where `pyarrow` or `fastparquet` is available and CSV otherwise. Each representation is identified by a key derived from a SHA-256 hash of a canonical JSON payload listing the selected feature families, Morgan radius and fingerprint bit width, so distinct configurations never collide. A schema file records the representation payload and column order; a mismatch raises an error rather than silently mixing definitions. Feature rows are keyed by canonical SMILES, so only molecules absent from the cache are computed on a given run, and cached and new rows are merged back into the exact requested row order. Additional caches store assembled benchmark feature matrices and tuned conventional models.

For TDC datasets, official splits were preferred over generic resampling. Where PyTDC exposes an official `admet_group` entry, the runner uses that `train_val/test` split in preference to legacy single-prediction cache entries; official split frames are cached under `data/_qsarena_cache` and stale entries lacking an official split are refreshed automatically. This distinction matters for interpretation and we preserve it throughout: 22 TDC datasets and 5 Polaris datasets used official predefined splits, while the remaining 17 used locally generated scaffold, random or target-quartile splits and are therefore *not* directly leaderboard-equivalent (main-text Section 3.5).

Four splitting strategies were supported — random, target-quartile, scaffold and predefined — selected per dataset with a benchmark-aware default, with a test fraction of 0.2 and random seed 13. Scaffold splits grouped molecules by stereochemistry-free Bemis–Murcko scaffold [8] and assigned whole scaffold groups to the test set by a greedy, seeded procedure, so that no scaffold appears in both partitions. Target-quartile splits stratified on quartiles of the target, reverting to random splitting when a dataset could not populate at least two bins per fold. Official TDC partitions were used directly, with deterministic reassignment by SMILES so that no molecule appears in both train and test. Cross-validation mirrored the split geometry (`GroupKFold` over scaffold groups, `StratifiedKFold` over quartile bins, shuffled `KFold` otherwise), with a defined fallback to random folds when group or bin constraints could not be met.

Before model fitting, a leak-free pre-filter removed degenerate and redundant columns using statistics computed on the training partition only and applied identically to the held-out set: near-constant columns below a variance threshold of  $1 \times 10^{-8}$  were dropped, binary columns whose positive prevalence fell outside 0.005–0.995 were removed, and exact-duplicate columns were pruned.

Selection was then performed train-only with a cross-validated elastic net. The `ElasticNetCV` selector was fit in an isolated subprocess with a wall-clock timeout over a configurable L1-ratio grid and

log-spaced alpha grid, using folds matched to the active split strategy. Features were retained when the absolute coefficient exceeded a threshold, with at least one feature always kept and the retained set capped at the larger of one feature or 10% of the training-row count. To bound runtime, the selector used a dataset-size runtime model: when predicted elastic-net runtime exceeded a configurable threshold (default 7200s), or when the elastic net timed out or failed, selection fell back to RandomForest importance ranking. Selected features, coefficients and diagnostics were written per dataset.

**Conventional machine learning.** For regression: elastic-net regression with internal cross-validated alpha/L1 search, support vector regression ( $C = 10$ ,  $\epsilon = 0.1$ , RBF kernel), random forests (400 trees), extremely randomized trees (500 trees), histogram gradient boosting (learning rate 0.05, up to 500 iterations, maximum depth 8), a soft-voting KNN+SVR regressor with adaptively chosen neighbour count, AdaBoost (500 estimators, learning rate 0.05), and a tabular multilayer perceptron (hidden layers 512 and 256, ReLU, Adam, L2  $1 \times 10^{-4}$ , up to 300 iterations), all via scikit-learn [9]. Gradient-boosting libraries were added when installed: XGBoost (400 trees, depth 6, learning rate 0.05, subsample and column-sample 0.9) [10], LightGBM (500 trees, learning rate 0.05, 63 leaves) [11] and CatBoost (400 iterations, depth 6, learning rate 0.05) [12]. Numeric pipelines were preceded by median imputation and standardization where appropriate. The classification table substituted the analogous estimators (logistic regression, SVC with probability estimates, random forest, extra trees, histogram gradient boosting, soft-voting KNN+SVC, AdaBoost, tabular MLP and the gradient-boosting classifiers) with classification losses and metrics. One further gradient-boosting model, XGBoost on the full, unselected ADMETboost feature set (MACCS keys, ECFP4 and PubChem fingerprints, Mol2Vec embeddings, and Mordred 2D and RDKit 2D descriptors; 1,000 trees, learning rate 0.05, maximum depth 6, row and column subsampling 0.8 and 0.5), bypasses feature selection altogether. It joined the library after the main run (main-text Section 2.12).

**Specialized tabular and graph models.** A compact one-dimensional convolutional regressor treats each standardized feature vector as a 1D signal and applies two same-padded ReLU convolutional blocks (64 filters, kernel size 5), global max pooling, a 128-unit dense layer with dropout and a linear output, trained with Adam and early stopping; it is intentionally small enough to run on CPU. The MapLight + GNN workflow fits a CatBoost model on the union of MapLight classic features and pretrained graph isomorphism network (GIN) fingerprints where the supporting DGL/PyTorch stack is available; a strict leaderboard-parity variant of the MapLight CatBoost model uses mean-absolute-error optimization, target scaling and five-seed averaging to reproduce the published protocol [5]. Graph neural networks were provided through Chemprop v2 [13, 14, 15] with configured directed message-passing (D-MPNN), CMPNN-style and AttentiveFP-style architectures; optional variants augment the graph encoder with train-only selected tabular descriptors or an RDKit-2D featurizer. Uni-Mol V1 and V2 [16, 17] were included as 3D pretrained baselines, running automatically when a GPU was detected. The configuration that produced the deposited results was 20 epochs, learning rate  $1 \times 10^{-4}$ , batch size 32, early stopping patience 5, a random internal validation split, a 64-atom cap and the 84m checkpoint, with automatic mixed precision enabled under an otherwise fp32 precision mode. These values are recorded in `run_config.json` and in the per-model `unimol_*` columns of every `metrics.csv`. The TabPFN tabular foundation model [18] was available for regression and classification; most of its results came from the Prior Labs API and some refits from the locally installed package (Note S1).

After base models produced aligned train and test prediction vectors, QSARena optionally fused them by combinatorial fusion analysis (CFA) [19] in both score and rank spaces. To control combinatorial growth, inputs were first reduced to the best model per workflow and the subset search bounded by a budget guardrail. For each candidate subset the algorithm computed a performance strength as the inverse of the base model’s training error and a diversity strength from the mean pairwise distance between normalized, sorted score profiles, then evaluated three score-space weightings (equal, performance-weighted, diversity-weighted) plus the corresponding rank-space variants. Rank-space combinations were linearly calibrated back to the target scale and given a small metric discount when subset diversity exceeded a threshold, so that diverse rank fusions were preferred only when genuinely complementary. Candidates were ranked by the adjusted training metric and the best fused predictor

returned with its selected models, weights and a candidate-diagnostics table. For regression, fusion minimized mean absolute error.

Three further ensemble strategies were available over the same prediction pool: out-of-fold stacking with a RidgeCV meta-model (logistic for classification), a weighted average with weights proportional to each member’s inverse out-of-fold error, and a simple average. Two optional member filters read the same out-of-fold predictions as member selection and weighting: one removes members with non-positive out-of-fold  $R^2$ , the other drops one member of each highly correlated pair. CFA outputs and members without out-of-fold predictions are excluded and listed in each ensemble’s notes.

Table S12 lists every model in the library with its per-dataset availability.

Table S12: Model inventory. “Valid” datasets are those on which the model produced a metric without an error; differences from 44 reflect task-type applicability (regression-only or classification-only estimators), dataset-size guardrails, or backend failures (main-text Section 3.9).

| Model family                    | Model                                                   | Valid regression datasets | Valid classification datasets | Wins | Datasets attempted |
|---------------------------------|---------------------------------------------------------|---------------------------|-------------------------------|------|--------------------|
| CFA combinatorial fusion        | CFA (Combinatorial Fusion)                              | 22                        | 22                            | 1    | 44                 |
| Chemprop v2 GNN                 | Chemprop v2 (AttentiveFP, ensemble=3)                   | 20                        | 18                            | 4    | 44                 |
| Chemprop v2 GNN                 | Chemprop v2 (CMPNN, ensemble=3)                         | 22                        | 20                            | 0    | 44                 |
| Chemprop v2 GNN                 | Chemprop v2 (D-MPNN + RDKit2D, ensemble=3)              | 20                        | 18                            | 1    | 44                 |
| Chemprop v2 GNN                 | Chemprop v2 (D-MPNN + Selected descriptors, ensemble=3) | 22                        | 20                            | 0    | 44                 |
| Chemprop v2 GNN                 | Chemprop v2 (D-MPNN, ensemble=3)                        | 22                        | 20                            | 0    | 44                 |
| Conventional ML                 | AdaBoost                                                | 22                        | 22                            | 0    | 44                 |
| Conventional ML                 | CatBoost                                                | 22                        | 22                            | 1    | 44                 |
| Conventional ML                 | ElasticNetCV                                            | 20                        | 0                             | 0    | 22                 |
| Conventional ML                 | Extra trees                                             | 22                        | 22                            | 0    | 44                 |
| Conventional ML                 | HistGradientBoosting                                    | 22                        | 22                            | 1    | 44                 |
| Conventional ML                 | LogisticRegression                                      | 0                         | 22                            | 0    | 22                 |
| Conventional ML                 | MapLight CatBoost (Strict Parity)                       | 22                        | 0                             | 0    | 22                 |
| Conventional ML                 | Random forest                                           | 22                        | 22                            | 1    | 44                 |
| Conventional ML                 | SVC                                                     | 0                         | 22                            | 0    | 22                 |
| Conventional ML                 | SVR                                                     | 22                        | 0                             | 0    | 22                 |
| Conventional ML                 | Tabular CNN                                             | 22                        | 0                             | 0    | 22                 |
| Conventional ML                 | Tabular MLP                                             | 22                        | 22                            | 0    | 44                 |
| Conventional ML                 | Voting Classifier (KNN, SVM)                            | 0                         | 22                            | 0    | 22                 |
| Conventional ML                 | Voting Regressor (KNN, SVM)                             | 22                        | 0                             | 0    | 22                 |
| Conventional ML                 | XGBoost                                                 | 22                        | 22                            | 0    | 44                 |
| Conventional ML                 | XGBoost (ADMETboost features)                           | 22                        | 22                            | 3    | 44                 |
| Deep tabular NN (ChemML MLP)    | ChemML MLP (PyTorch)                                    | 22                        | 22                            | 0    | 44                 |
| Deep tabular NN (ChemML MLP)    | ChemML MLP (TensorFlow)                                 | 22                        | 22                            | 0    | 44                 |
| Ensemble (stacking / averaging) | Ensemble (OOF Stacking (RidgeCV, 5-fold))               | 22                        | 22                            | 18   | 44                 |
| Ensemble (stacking / averaging) | Ensemble (Weighted average (inverse OOF error))         | 22                        | 22                            | 4    | 44                 |
| MapLight + GNN                  | MapLight + GNN (CatBoost, Strict Parity)                | 22                        | 22                            | 2    | 44                 |
| TabPFN (tabular foundation)     | TabPFNClassifier                                        | 0                         | 19                            | 0    | 22                 |
| TabPFN (tabular foundation)     | TabPFNRegressor                                         | 22                        | 0                             | 3    | 22                 |
| Uni-Mol (3D pretrained)         | Uni-Mol V1                                              | 22                        | 22                            | 2    | 44                 |
| Uni-Mol (3D pretrained)         | Uni-Mol V2 (84m)                                        | 10                        | 7                             | 3    | 44                 |

## Supplementary Note S3: Sensitivity of the family comparison

Both analyses in this note read only the committed per-dataset metric tables of the manuscript run; no model was retrained. They are produced by `reanalysis_coverage_posthoc.py`, which first reproduces every family’s best model per dataset exactly as the main analysis reports it, and their numbers are checked against the manuscript by the verification script.

**Common-subset comparison.** Model coverage is uneven (Table S4): Chemprop v2 produced valid results on 38 to 42 of the 44 datasets depending on variant, TabPFN on 41, and Uni-Mol V2 on 17 (Uni-Mol V1 on all 44, so the Uni-Mol family has a result everywhere). Table S13 recomputes each family’s wins, median relative gap to the per-dataset best and share of datasets within 5% of the best on all 44 datasets, on the 41 datasets on which all eight families produced a result, and on the 17 datasets on which Uni-Mol V2 ran. Restricting the dataset set does not change any dataset’s winner, which is chosen over all models on that dataset; it changes which datasets are counted. The single-family win counts are unchanged on the 41-dataset subset, and conventional machine learning keeps the smallest median gap of any single family on both subsets, so the comparison between conventional and pretrained families does not depend on which datasets each family covered.

**Post-hoc descriptor model.** XGBoost on the full, unselected ADMETboost feature set joined the library after it performed well in an exploratory analysis (main-text Section 2.12). Adding a model after seeing results can flatter test-selected comparisons, so Table S14 recomputes the win tally with that model removed as a candidate. It won three datasets; without it, two of them go to the out-of-fold stacking ensemble and one to Uni-Mol V2, so conventional machine learning falls from 6 wins to 3, and no single family wins more than 6. The ensemble rows still contain the descriptor model as a member on the datasets where it qualified; removing it there would need an ensemble rebuild, which was not done, so the table bounds only its direct effect.

**Exploratory analysis behind the descriptor model.** The descriptor model came from an exploratory analysis run after the main benchmark: fixed-configuration XGBoost on the full, unselected ADMETboost feature set (main-text Section 2.6). In the benchmark it won 3 of the 44 datasets outright (CYP3A4 substrate, HIA and skin reaction). Because it selects no features, its cross-validation scores never needed the nested selection of main-text Section 2.5. Scored on its own against the 22 official TDC splits, it beat the published ADMETboost results on 12 of the 22 datasets and the NIST meta-model on 13, but lost to MaxQsaring on 17. In that exploratory analysis, adding label-free pretrained embeddings (CheMeleon and Uni-Mol) did not improve it: across the 35 datasets with a comparable test metric the median change was  $-0.04\%$  (Wilcoxon  $p = 0.30$ ), and the embeddings alone were worse (median  $-2.42\%$ ,  $p < 0.001$ ). A Chemprop variant initialised from CheMeleon is available in QSARena as an opt-in model and was not used in the benchmark. The analysis is deposited in `benchmark_results/qsarena_feature_expansion/`.

Table S13: Common-subset sensitivity of the family comparison: wins, median relative gap of each family’s best model to the per-dataset best, and share of datasets within 5% of the best, on all datasets, on the datasets on which all eight families produced a valid result, and on the datasets on which Uni-Mol V2 produced one. Winners are chosen over all models on each dataset, so restricting the dataset set changes which datasets are counted, not any dataset’s winner. Single split and single seed.

| Dataset subset        | Model family                    | Datasets | Wins | Median gap to best (%) | Within 5% of best (%) |
|-----------------------|---------------------------------|----------|------|------------------------|-----------------------|
| All datasets (n = 44) | Ensemble (stacking / averaging) | 44       | 22   | 0.0                    | 84.0                  |
| All datasets (n = 44) | Conventional ML                 | 44       | 6    | 3.2                    | 66.0                  |
| All datasets (n = 44) | Chemprop v2 GNN                 | 42       | 5    | 5.6                    | 48.0                  |
| All datasets (n = 44) | Uni-Mol (3D pretrained)         | 44       | 5    | 5.4                    | 48.0                  |
| All datasets (n = 44) | TabPFN (tabular foundation)     | 41       | 3    | 8.4                    | 22.0                  |
| All datasets (n = 44) | MapLight + GNN                  | 44       | 2    | 12.5                   | 11.0                  |

*continued on next page*

| Dataset subset                    | Model family                    | Datasets | Wins | Median gap<br>to best<br>(%) | Within 5%<br>of best<br>(%) |
|-----------------------------------|---------------------------------|----------|------|------------------------------|-----------------------------|
| All datasets (n = 44)             | CFA combinatorial fusion        | 44       | 1    | 5.2                          | 50.0                        |
| All datasets (n = 44)             | Deep tabular NN (ChemML MLP)    | 44       | 0    | 16.6                         | 18.0                        |
| All eight families valid (n = 41) | Ensemble (stacking / averaging) | 41       | 19   | 0.1                          | 83.0                        |
| All eight families valid (n = 41) | Conventional ML                 | 41       | 6    | 3.7                          | 63.0                        |
| All eight families valid (n = 41) | Chemprop v2 GNN                 | 41       | 5    | 5.9                          | 46.0                        |
| All eight families valid (n = 41) | Uni-Mol (3D pretrained)         | 41       | 5    | 5.8                          | 44.0                        |
| All eight families valid (n = 41) | TabPFN (tabular foundation)     | 41       | 3    | 8.4                          | 22.0                        |
| All eight families valid (n = 41) | MapLight + GNN                  | 41       | 2    | 12.8                         | 10.0                        |
| All eight families valid (n = 41) | CFA combinatorial fusion        | 41       | 1    | 5.8                          | 46.0                        |
| All eight families valid (n = 41) | Deep tabular NN (ChemML MLP)    | 41       | 0    | 17.7                         | 12.0                        |
| Uni-Mol V2 valid (n = 17)         | Ensemble (stacking / averaging) | 17       | 8    | 0.2                          | 76.0                        |
| Uni-Mol V2 valid (n = 17)         | Conventional ML                 | 17       | 2    | 3.7                          | 53.0                        |
| Uni-Mol V2 valid (n = 17)         | Chemprop v2 GNN                 | 17       | 1    | 7.6                          | 47.0                        |
| Uni-Mol V2 valid (n = 17)         | Uni-Mol (3D pretrained)         | 17       | 4    | 5.4                          | 47.0                        |
| Uni-Mol V2 valid (n = 17)         | TabPFN (tabular foundation)     | 16       | 2    | 9.5                          | 31.0                        |
| Uni-Mol V2 valid (n = 17)         | MapLight + GNN                  | 17       | 0    | 13.9                         | 12.0                        |
| Uni-Mol V2 valid (n = 17)         | CFA combinatorial fusion        | 17       | 0    | 5.9                          | 35.0                        |
| Uni-Mol V2 valid (n = 17)         | Deep tabular NN (ChemML MLP)    | 17       | 0    | 20.1                         | 12.0                        |

Table S14: Datasets won by each family with and without the post-hoc descriptor model (XGBoost on the full, unselected ADMETboost feature set) as a candidate. The ensemble rows are unchanged models that still contain it as a member; removing it there would need an ensemble rebuild, so the table bounds only its direct effect.

| Model family                    | Wins with the<br>descriptor<br>model | Wins without it | Change |
|---------------------------------|--------------------------------------|-----------------|--------|
| Ensemble (stacking / averaging) | 22                                   | 24              | 2      |
| Conventional ML                 | 6                                    | 3               | -3     |
| Chemprop v2 GNN                 | 5                                    | 5               | 0      |
| Uni-Mol (3D pretrained)         | 5                                    | 6               | 1      |
| TabPFN (tabular foundation)     | 3                                    | 3               | 0      |
| MapLight + GNN                  | 2                                    | 2               | 0      |
| CFA combinatorial fusion        | 1                                    | 1               | 0      |
| Deep tabular NN (ChemML MLP)    | 0                                    | 0               | 0      |

## Supplementary Note S4: When does each family win? A dataset-property meta-analysis

Section 3.2 of the main text found that no model family dominates. Here we ask when each family comes close to the best, using properties of the dataset alone. This is per-dataset algorithm selection from dataset meta-features, the approach Meta-QSAR applied to thousands of QSAR problems [20], and it addresses the observation that the best ADMET model and representation are strongly dataset-dependent [21]. For each of the 44 datasets we computed meta-features on the exact train/test partition used in the benchmark, verified against the recorded split hashes (Table S9). They are: training-set size, label imbalance or skew, Bemis–Murcko scaffold diversity, internal fingerprint diversity, and train-to-test similarity. Similarity is summarised by SNN, the Tanimoto similarity of each test molecule to its nearest training molecule, which is known to track prediction error [22]. Winners are single-split, single-seed outcomes, so the outcome modelled is the continuous relative gap of Figure S5 (Note S6), never the identity of the winner. Every interval is a 95% percentile bootstrap over datasets.

**Training-set size.** The gap of the best conventional-ML model to the per-dataset best showed no clear monotone trend with training-set size (Spearman  $\rho = 0.03$ , 95% CI  $-0.29$  to  $0.35$ , BH  $q = 0.87$ ). For the pre-registered contrast between conventional ML and the 3D-pretrained Uni-Mol family (Figure S1), the fitted gap curves do not cross inside the observed size range (a crossing occurred in only 30% of dataset-bootstrap replicates). Tuned comparisons of D-MPNN and random forest report crossovers at roughly 500–2,000 training compounds [23], and dataset size is known to govern when representation learning pays off [24]. Under QSARena’s single fixed configuration, no such crossover is resolved: neither family’s fitted gap overtakes the other within the observed sizes.

**Chemical-space shift.** Mean SNN was not clearly associated with the achievable best held-out metric (within-task rank; Spearman  $\rho = 0.17$ , 95% CI  $-0.14$  to  $0.48$ , permutation  $p = 0.262$ ; Figure S2a). Of the screened properties, the out-of-domain fraction (test molecules with SNN  $< 0.40$ ) had the strongest association with achievable performance ( $\rho = -0.26$ , 95% CI  $-0.57$  to  $0.08$ ), although its interval also includes zero. Message-passing models have been reported to generalise to unseen chemical space better than tree-based models [13], and the 7 datasets re-split to scaffold splits between our two runs (main-text Section 3.10) offer a small natural experiment on identical chemistry (Figure S2b). Moving them to scaffold splits changed the median family gap by +1.8 percentage points for conventional ML and +1.1 for Uni-Mol. Families valid in both runs on fewer than 3 of them are not summarised: Chemprop (1 of 7). With 7 datasets, and with the two runs also differing in hardware and model settings, this is descriptive only.

**Family  $\times$  property grid.** Across 48 family  $\times$  meta-feature correlations with Benjamini–Hochberg control (Table S10), one survived at  $q < 0.10$ , the strongest being Ensemble (stacking / averaging) versus log10 training-set size ( $\rho = -0.47$ , 95% CI  $-0.70$  to  $-0.19$ ,  $q = 0.06$ ).

**A family recommender.** A leave-one-dataset-out depth-3 decision tree used four meta-features: log10 training-set size, mean SNN, label asymmetry and task. It predicted the best family group (fusion, descriptor-based ML, Uni-Mol or Chemprop) with balanced accuracy 0.32 (95% CI 0.23 to 0.41). The majority-class baseline scored 0.25, and label permutation scored 0.24 (permutation  $p = 0.138$ ; Figure S3). Its interval overlaps the baselines, so dataset meta-features alone do not reliably identify the winning family at this sample size; the per-dataset winner remains an empirical question, which is the case for benchmarking many families on every dataset.

Predicting the winner’s identity is a harsh test, because many datasets have several families within a few percent of each other. Algorithm selection is usually judged instead by regret, the performance given up by the recommended choice [20]. Here regret is the gap of the recommended family, and the selector was designed and fixed before any result was seen. Each held-out dataset’s selector was chosen by an inner leave-one-dataset-out loop among 12 variants. These were nearest-dataset and per-family ridge models over size, similarity, chemistry, label-landscape and training-set cross-validation landmark features. Nested selection did not reliably beat the single best family (mean regret 3.1% versus 3.3% for always choosing the family with the best average record (difference 0.2 percentage points, 95% CI  $-0.9$  to  $1.3$ ; permutation  $p = 0.099$ )). Both lose little: a randomly chosen valid family would cost 10.7% on average. The picked family was within 5% of the best on 86% of datasets, against 84% for

the single best family. The inner loop most often chose the nearest-datasets model on F0+F3 (66% of held-out datasets; all variants in Table S11).

Diversity metrics depend on the fingerprint. Recomputed with ECFP6 at 4,096 bits, the fingerprint-dependent correlations kept their sign in 92% of cells and their significance classification in 100%. These relationships are exploratory. They rest on one split and one seed per dataset and on  $n = 44$  datasets, so they generate hypotheses for a multi-seed or learning-curve study; they do not establish selection rules.

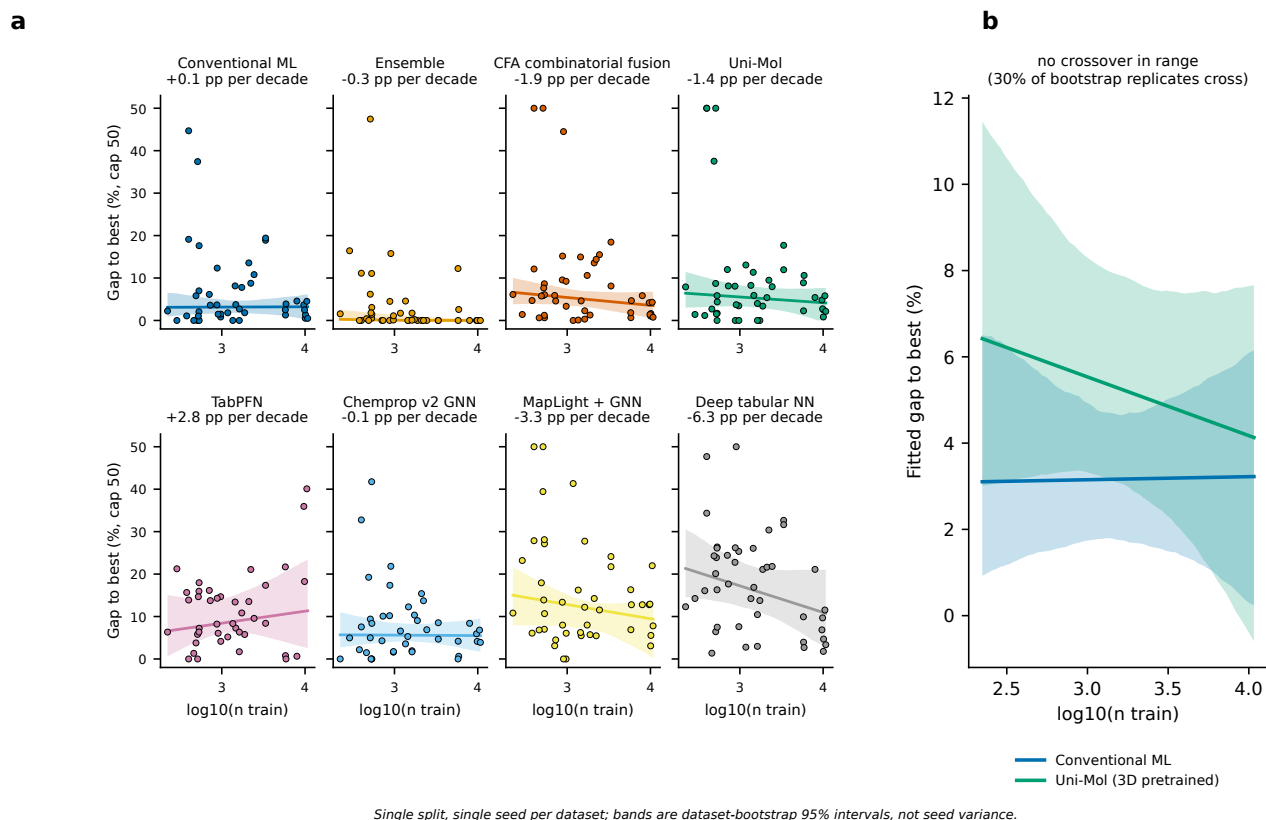


Figure S1: (a) Relative gap of each family's best model to the per-dataset best versus log10 training-set size, with Theil–Sen fits and dataset-bootstrap 95% bands (gaps capped at 50% for display). (b) Fitted gap curves for conventional ML and Uni-Mol; the dashed line and grey span mark the estimated crossover and its bootstrap CI when one exists. Single split and single seed per dataset: bands reflect dataset resampling, not seed variance.

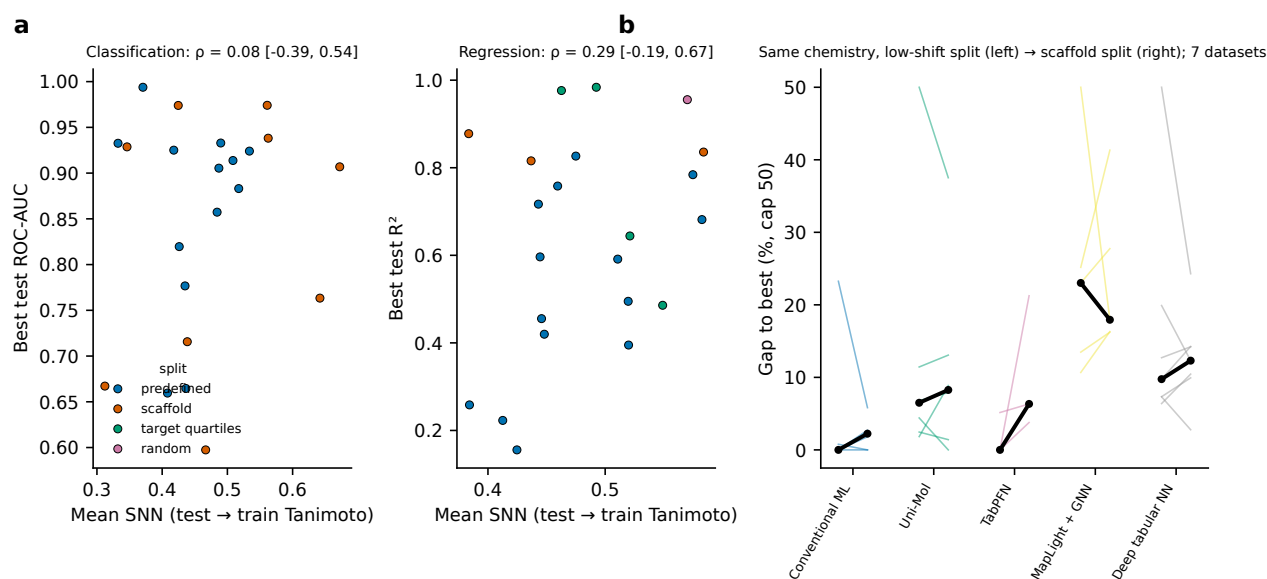


Figure S2: (a) Achievable best held-out metric versus mean test-to-train nearest-neighbour Tanimoto (SNN), by task and split protocol. (b) Family gaps on the 7 datasets that were re-split from random or target-quartile splits to scaffold splits between the RTX 4060 and A100 runs (main-text Section 3.10); thin lines are datasets, thick lines are medians.

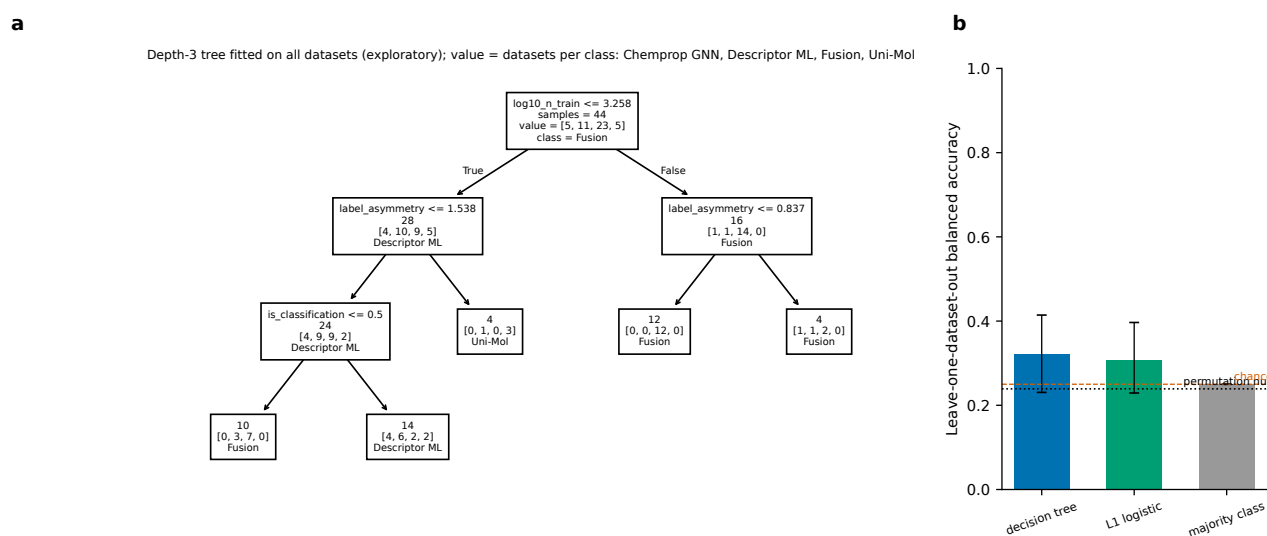


Figure S3: (a) Depth-3 decision tree over four meta-features, fitted on all datasets for display. (b) Leave-one-dataset-out balanced accuracy of the tree and an L1-penalised multinomial logistic model against the majority-class baseline (grey), the label-permutation null (dotted) and chance (dashed), with dataset-bootstrap 95% CIs. Exploratory:  $n = 44$ .

## Supplementary Note S5: Comparison with existing automated QSAR platforms

This note gives the full comparison that main-text Section 3.12 summarizes.

Automating the QSAR model-building loop is not a new idea. The main-text Introduction distinguished QSARena from DeepChem, QSPRpred, OCHEM, ChemSAR and ADMET-AI; here we compare it with the automated QSAR systems closest to it in purpose.

**QSAR Workbench** [25] is the closest intellectual ancestor. Built on Pipeline Pilot, it systematically explores combinations of descriptor sets, learners and train/test splits, generating on the order of 600 models in under an hour, and it reports performance equivalent to published state-of-the-art approaches on its demonstration data. Its authors are explicit that the system has no built-in competitive model selection and that choosing among the generated models requires human judgement. That is precisely the step QSARena automates and then audits: the per-dataset winner is selected programmatically, and main-text Section 3.5 quantifies how much of the resulting apparent performance is an artifact of making that selection on held-out data. QSAR Workbench is commercial, depends on a proprietary workflow engine, and was demonstrated on two datasets (209 and ~670 compounds); QSARena is MIT-licensed, depends only on open Python packages, and is evaluated here on 44 datasets spanning 280 to 13,445 molecules.

**Schrödinger AutoQSAR** [26] and its successor DeepAutoQSAR occupy the same application area as this work. AutoQSAR automates model building, validation and deployment, runs on Windows, macOS and Linux through either a command line or the Maestro GUI, and is distributed as closed-source software within the commercial Schrödinger Suite. Schrödinger has published a benchmark of DeepAutoQSAR against ChemProp and DeepPurpose on the ADMET subset of the Therapeutics Data Commons [27], reporting that DeepAutoQSAR ranked among the top performers on 20 of 22 cases and clearly outperformed the comparators on 9 of them.

A direct accuracy comparison against that study is not possible, and we do not attempt one. The white paper defines “top performer” and “clearly outperforming” qualitatively rather than as a numerical criterion, does not report per-dataset metric values, and does not release per-molecule predictions, so its results cannot be re-scored under our protocol or ours under theirs. What can be compared is summarized in Table S15, and it is less than it first appears. DeepAutoQSAR’s “top performer on 20 of 22” is its standing against two named comparators, ChemProp and DeepPurpose, not a position on the TDC leaderboard; QSARena’s 22 of 22 is a rank among curated published reference values. The two numbers are not commensurable. What the two systems can be compared on is every axis that governs who can actually use them, and there they differ sharply.

Three differences deserve emphasis. On **cost and access**, the Schrödinger tools require a commercial licence and QSAR Workbench requires a Pipeline Pilot licence, whereas QSARena is free, MIT-licensed, installable with `pip`, and runnable without any installation at all in a hosted notebook (main-text Section 3.11). On **transparency**, the comparators’ benchmark evidence is a vendor white paper and a demonstration study respectively; QSARena’s is a deposited artifact trail from which every number in this paper regenerates, together with a script that fails if the text and the artifacts disagree. On **breadth**, both prior systems are single-suite; QSARena is evaluated across five.

We also note what the commercial tools plausibly do better. Neither the QSAR Workbench nor the Schrödinger products carry the coverage gaps we report in main-text Section 3.9, both are professionally supported, and DeepAutoQSAR’s benchmark reflects sustained engineering investment in a single well-tuned engine; ours is a broad but shallow sweep. A reader whose priority is a supported production pipeline on one endpoint class may reasonably prefer them; a reader who needs auditability, cross-suite breadth, or zero cost should prefer this work.

**MetaQSAR** [28] is a third, more recent system and the closest comparator for a regulatory audience. It is a standalone Java desktop application, free to download after registration, with fully automated and manual modes. It computes PaDEL descriptors and fingerprints, selects features by genetic algorithm, stepwise and filter methods, fits multiple linear regression, partial least squares, linear discriminant analysis, random forest and support-vector models, and ships two applicability-domain methods — a descriptor standardization approach [29] and a posterior-probability confidence

approach for classifiers — with explicit alignment to OECD and ECHA good-practice guidance. Its published evidence is four endpoint case studies, mostly environmental toxicity, in which its models matched or exceeded previously reported QSAR models on the same data, rather than public leaderboards. QSARena is complementary: where MetaQSAR is GUI-first, classical-descriptor-based and case-study-validated, QSARena is programmatic and code-free through a hosted notebook, spans deep, 3D-pretrained and fusion architectures, and is evaluated uniformly across five public benchmark suites. MetaQSAR’s regulatory framing is the one QSARena previously lacked, and main-text Section 3.13 adopts it.

**Relationship to ADMET-AI, stated precisely.** ADMET-AI is the reference open tool in this space. It is *not* one of the 31 models evaluated here; it appears only as a cited comparator. What QSARena integrates is the package ADMET-AI is built on: our **Chemprop v2** (D-MPNN + RDKit2D) variant concatenates a Chemprop graph representation with RDKit physicochemical descriptors, which is essentially ADMET-AI’s recipe, but it is a reimplementations through Chemprop v2 rather than ADMET-AI’s own code, package or trained weights.

That path remained incomplete in this run. The ADMET-AI-like Chemprop v2 (D-MPNN + RDKit2D) variant produced valid results on 38 of 44 datasets, so it is informative but still not a head-to-head reproduction of ADMET-AI. **We therefore cannot claim to reproduce or to beat ADMET-AI; parity is untested.** We ran no head-to-head comparison against it, and none should be inferred from the estimated ranks reported here.

**Where QSARena does not lead.** Published rankings are almost always each paper’s rank *at its own publication date*. Our curated reference table inherits that convention: it contains 64 rows claiming first place across 28 datasets, and **caco2\_wang** alone carries four different models recorded as rank 1. Such figures cannot be compared with one another, and none of MaxQsaring, ADMETboost, ADMET-AI, Auto-ADMET or DeepAutoQSAR appears in the TDC leaderboard snapshots we captured, so no head-to-head measurement against them exists.

We therefore re-scored every reference value in the honest direction for its metric, producing one consistent ranking per dataset. This is a materially different picture from the published claims. MaxQsaring, whose paper reports first place on 19 of 22 TDC tasks [30], holds 7 first places under joint re-scoring but is top-3 on 19 of 22 with a median rank of 2 — the strongest system in the reference set, and the fairer way to state its lead. ADMETboost’s reported 18 first places [31] reduce to **zero**: it was genuinely first in 2022 and has since been overtaken on all 22 datasets, which is a caution about citing leaderboard positions as though they were durable. MapLight’s top-3 placement on 16 of 22 [5] is an author self-report we cannot re-score, and DeepAutoQSAR’s “top performer on 20 of 22” [27] is best-of-three against ChemProp and DeepPurpose in a vendor white paper rather than a leaderboard position at all — the two figures were never in competition. We also corrected three reference values that were on the wrong scale, all from a model reporting normalised targets (for example a PPBR MAE of 0.679 against a leaderboard best of 7.440).

Against that corrected reference set, QSARena’s best model per dataset reaches a median rank of 2.5 on the 22 official TDC tasks under test-based selection, and falls to a median rank of 7 under cross-validation-only selection (main-text Section 3.5). MaxQsaring’s median rank of 2 is therefore better than ours under either protocol. We make no claim to accuracy leadership. The contribution is the cross-suite benchmark and the selection-protocol decomposition; code-free access is a usability feature.

**Per-dataset placements.** Figure S4 shows, for each of the 37 comparable datasets, the estimated rank of the test-selected best model and of the cross-validation-selected model (values in Table S7; main-text Section 3.5 and Figure 3 give the distribution).

The same landscape shows that rigorous benchmarking builds credibility but does not by itself win users. The most widely used tools in this space — SwissADME [32], pkCSM [33] and ADMETlab [34] — are free, fixed-model web services that do not report on the TDC leaderboard at all, and they are popular because they are convenient. QSARena aims to offer both, and we report the two claims separately.

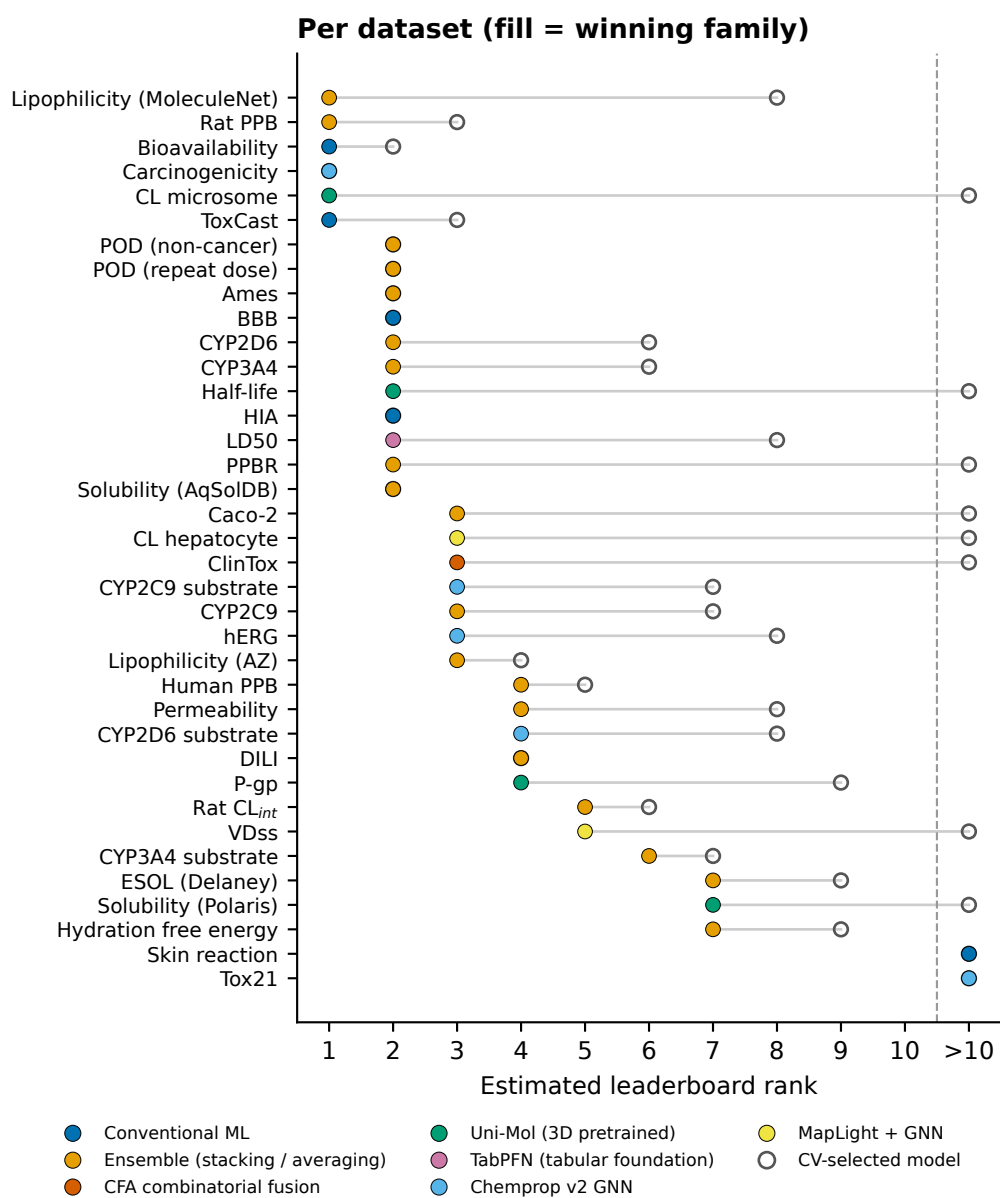


Figure S4: Estimated leaderboard rank per dataset across the 37 comparable datasets. Filled markers are the test-selected best model, coloured by model family; open markers are the model selected by cross-validation alone. The dashed line marks the top-ten boundary. Single split and single seed: individual placements are provisional and no variance is estimated.

Table S15: Positioning of QSARena relative to representative open-source, commercial and web-based molecular property-prediction tools. Performance entries report each tool’s **own published results** and are **not** head-to-head comparisons: rows use different splits, seeds and selection protocols, and “—” indicates no public TDC leaderboard results. Where a published claim could be re-scored against the full reference set we give both, because they differ substantially: ADMETboost’s 18 reported first places become 0 once all reference values are ranked together, and MaxQsaring’s 19 become 7. None of the tools in this table appears in the TDC leaderboard snapshots we captured, so every performance entry is publication-reported. ADMET-AI’s “highest average rank” is a 2024 self-reported aggregate, not per-endpoint supremacy. QSARena reuses ADMET-AI’s Chemprop-RDKit architecture through its **Chemprop v2** (D-MPNN + RDKit2D) variant but does not integrate ADMET-AI itself, and in this run that variant produced valid results on 38 of 44 datasets, so parity with ADMET-AI remains untested. “Code-free” means usable without scripting; “re-train” means the user can fit models on their own data; “suites” counts distinct benchmark collections the tool has been evaluated on in its own publications.

| Tool                                         | Access / licence | Code-free               | Retrain                | Architecture breadth                                  | Suites                  | Published TDC ADMET performance                                                                                                 |
|----------------------------------------------|------------------|-------------------------|------------------------|-------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>QSARena (this work)</b>                   | Open, MIT        | Yes (Colab + CLI)       | Yes                    | 31 models, 8 families                                 | 5                       | Est. top-10 on 22/22 (test-selected); 16/22, median rank 7 under CV-only selection                                              |
| ADMET-AI [35]                                | Open, MIT        | Partial (web + CLI)     | Yes (training scripts) | 1 (Chemprop-RDKit)                                    | 1 (41 datasets)         | Highest average rank on TDC ADMET leaderboard (self-reported, 2024)                                                             |
| MaxQsaring [30]                              | Open, MIT        | No (Python + CLI)       | Yes                    | Multi-fingerprint feature selection + boosting        | 1 (22 datasets)         | Reported first on 19/22; <b>re-scored: 7 first places, top-3 on 19/22, median rank 2</b>                                        |
| ADMETboost [31]                              | Open             | No (Python)             | Yes                    | 1 (XGBoost on multi-fingerprint features)             | 1 (22 datasets)         | Reported first on 18/22 (2022); <b>re-scored: 0 first places, top-3 on 6/22</b>                                                 |
| Schrödinger AutoQSAR / DeepAutoQSAR [26, 27] | Commercial       | Yes (Maestro GUI + CLI) | Yes                    | Multi-method ML + GNN                                 | 1                       | Top of three on 20/22, clearly best on 9/22 — a three-way vendor comparison vs ChemProp and DeepPurpose, not a leaderboard rank |
| QSARtuna [36]                                | Open, Apache-2.0 | No (Python + CLI)       | Yes                    | Representations × learners via Optuna; + Chemprop; UQ | 0 (illustrative)        | —                                                                                                                               |
| ZairaChem [37]                               | Open, GPL-3.0    | No (CLI)                | Yes                    | AutoML ensemble + distillation                        | 1 (classification only) | State of the art out of the box on TDC classification                                                                           |
| QSPRpred [38]                                | Open             | No (Python library)     | Yes                    | Flexible descriptors × learners                       | 0 (framework)           | —                                                                                                                               |

| Tool                                 | Access / licence                         | Code-free           | Retrain    | Architecture breadth                                                 | Suites                      | Published TDC<br>ADMET performance           |
|--------------------------------------|------------------------------------------|---------------------|------------|----------------------------------------------------------------------|-----------------------------|----------------------------------------------|
| DeepChem [39]                        | Open, MIT                                | No (Python library) | Yes        | Many graph, transformer and classical models; many featurizers       | MoleculeNet loaders         | —                                            |
| OCHEM [40]                           | Web platform (registration)              | Yes (web)           | Yes        | Many descriptor sets $\times$ ML methods; applicability domain       | n/a                         | —                                            |
| ChemSAR [41]                         | Free web                                 | Yes (web)           | Yes        | Descriptors $\times$ scikit-learn learners                           | n/a                         | —                                            |
| DeepMol [42]                         | Open, Apache                             | No (Python)         | Yes        | AutoML over many models                                              | 1 (22 datasets)             | Competitive pipelines across 22 TDC datasets |
| DeepPurpose [43]                     | Open                                     | No (Python)         | Yes        | 15 encoders, 50+ models                                              | TDC loader                  | Used as TDC baselines                        |
| ChemXploreML [44]                    | Open                                     | Yes (desktop GUI)   | Yes        | Embeddings + tree ensembles                                          | 0 (physicochemical)         | —                                            |
| MetaQSAR [28]                        | Free (registration); standalone Java GUI | Yes (desktop GUI)   | Yes        | Classical: MLR, PLS, LDA, RF, SVM on PaDEL descriptors; 2 AD methods | 0 (4 endpoint case studies) | —                                            |
| QSAR Workbench [25]                  | Commercial (Pipeline Pilot)              | Yes (protocols)     | Yes        | Descriptor $\times$ learner grid                                     | 0 (2 CAESAR datasets)       | —                                            |
| ADMET Predictor (Simulations Plus)   | Commercial                               | Yes (GUI + CLI)     | Yes        | Proprietary, $\sim$ 100 endpoints                                    | Proprietary                 | —                                            |
| StarDrop / Auto-Modeller (Optibrium) | Commercial                               | Yes (GUI)           | Yes        | PLS, RF, SVM, GP                                                     | Proprietary                 | —                                            |
| SwissADME [32]                       | Free web                                 | Yes (web)           | No (fixed) | Fixed in-house models                                                | n/a                         | —                                            |
| pkCSM [33]                           | Free web                                 | Yes (web)           | No (fixed) | Graph-based signatures                                               | n/a                         | —                                            |
| ADMETlab 3.0 [34]                    | Free web                                 | Yes (web)           | No (fixed) | Multi-task graph attention                                           | Internal                    | —                                            |

## Supplementary Note S6: Coverage, cost and hardware details

**Model coverage and backend failures.** Not every model ran on every dataset, and we report the gaps rather than silently analysing only successes (Table S4). Three sources account for nearly all of them. Task applicability: regression-only and classification-only estimators are valid on at most 22 datasets each. Guardrails and capacity: Uni-Mol V2 produced valid results on 17 of 44 datasets at the 84 M-parameter size, and its 164 M and 310 M variants produced none, so the V2 results here are a partial view of that architecture. Backend failures: the Chemprop v2 variants were substantially repaired but still did not all finish everywhere, with 38 to 42 valid datasets per variant. TabPFN produced results on 41 of the 44 datasets; on the other three its label-based results were withdrawn (Note S1). Table S13 (Note S3) shows that restricting the family comparison to the datasets on which every family produced a result leaves it unchanged. Diagnosing the larger Uni-Mol V2 variants and completing Chemprop on its remaining datasets are the first things to fix before any follow-up benchmark. Figure S5 shows the relative gap of every family’s best model to the per-dataset best on every dataset, including the families with no valid result there.

**Cost.** Recorded model sizes underline the accessibility argument. The models that won most of our datasets — gradient-boosted trees and the linear meta-models over them — carry no pretraining corpus at all, against published comparators MolE ( $\sim 100$  M parameters, pretrained on  $\sim 842$  M molecules) [45] and MolGPS ( $\sim 3$  B parameters) [46]. Table S16 gives the full per-family cost table with these published comparators.

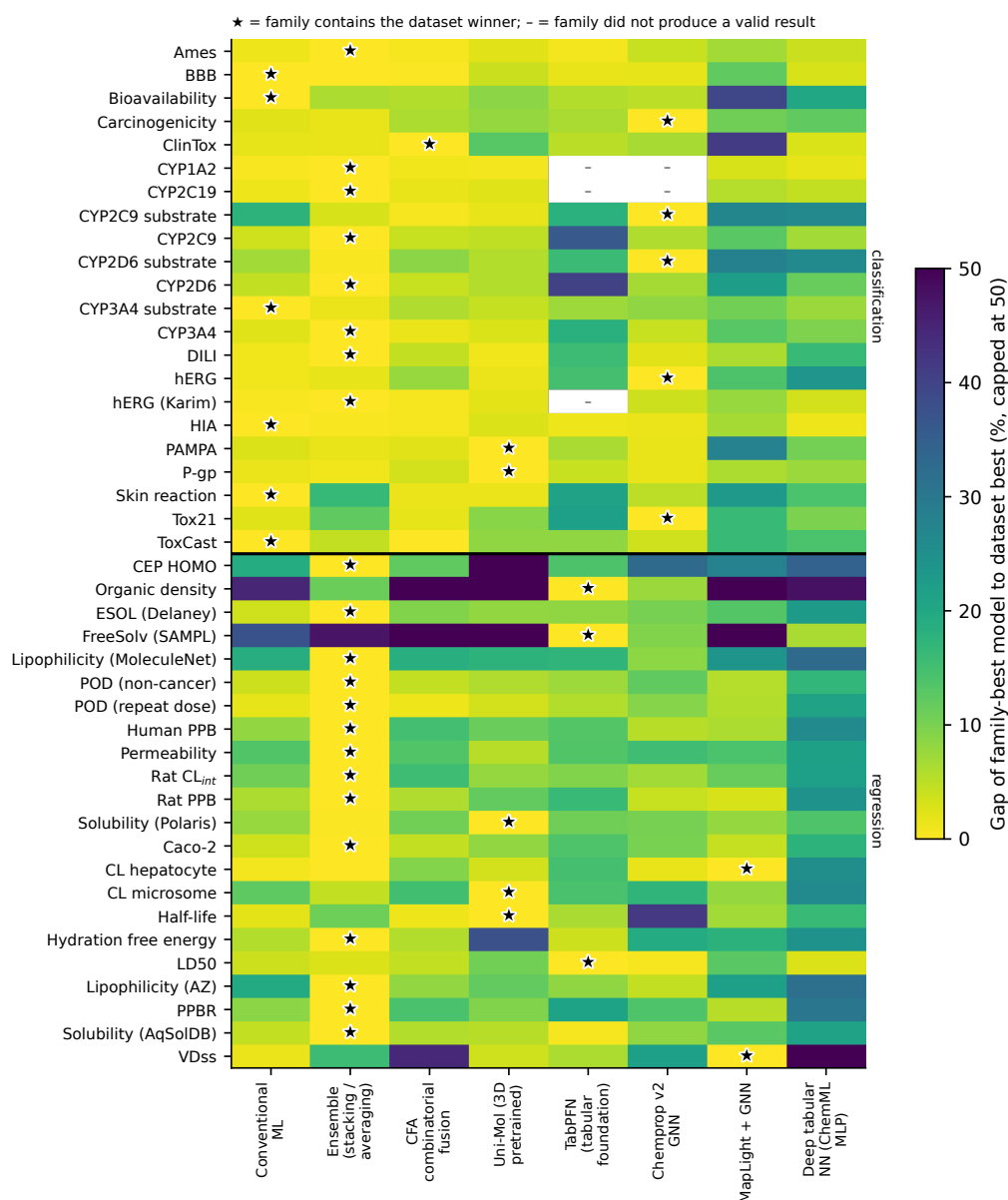


Figure S5: Relative gap of each family's best model to the per-dataset best (% , capped at 50), for every dataset. Stars mark the family containing the dataset winner; dashes mark families that produced no valid result for that dataset. Classification datasets are shown above the rule, regression below. Single split and single seed.

Table S16: Per-family computational cost and model size in this benchmark, with published comparators. Wall-clock times are per model per dataset on the run hardware.

| Model family                       | Model-dataset<br>fits timed | Median own<br>wall-clock<br>(s) | IQR own<br>wall-clock (s) | Median cost<br>incl. base pool<br>(s) | Median<br>trainable<br>parameters | Notes                                                                                             |
|------------------------------------|-----------------------------|---------------------------------|---------------------------|---------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------|
| CFA combinatorial<br>fusion        | 44.0                        | 0.3                             | 0-0                       | 1,378                                 |                                   | fusion over fitted base-model predictions; timing from canonical full run                         |
| Ensemble (stacking /<br>averaging) | 87.0                        | 0.6                             | 0-1                       | 1,378                                 |                                   | fusion over fitted base-model predictions; timing from canonical full run                         |
| Conventional ML                    | 440.0                       | 6.8                             | 3-20                      |                                       | 157,953                           | measured in canonical full run                                                                    |
| Deep tabular NN<br>(ChemML MLP)    | 88.0                        | 39.8                            | 27-94                     |                                       | 90,369                            | measured in canonical full run                                                                    |
| MapLight + GNN                     | 44.0                        | 137.4                           | 126-160                   |                                       |                                   | measured in canonical full run                                                                    |
| Chemprop v2 GNN                    | 25.0                        | 268.6                           | 229-274                   |                                       | 325,552                           | measured in canonical full run                                                                    |
| Uni-Mol (3D pretrained)            | 61.0                        | 373.5                           | 170-1285                  |                                       | 47,331,652                        | measured in canonical full run                                                                    |
| MolGPS (published)                 |                             |                                 |                           |                                       |                                   | ~3B parameters; GPU pretraining/inference reported in literature                                  |
| MolE (published)                   |                             |                                 |                           |                                       |                                   | ~100M parameters; GPU pretraining reported in literature                                          |
| ADMET-AI (published)               |                             |                                 |                           |                                       |                                   | Chemprop-RDKit; exact parameter count not recorded here; GPU-capable<br>Chemprop-RDKit deployment |

Feature selection scaled sub-linearly with dataset size in this run: a log-log fit of selector time against dataset size gives a slope of 0.77 (Pearson  $r = 0.58$  across 44 datasets), with a median of 295 s and a maximum of 1,003 s on `tdc_herg_karim`. This run disabled the dataset-size RandomForest fallback (main-text Section 2.5), so every dataset used the full elastic-net selector; the flatter scaling relative to our consumer-GPU run reflects 32-way parallelism rather than a change in the algorithm.

**Hardware comparison.** Because accessibility is a central claim of this work, we repeated the analysis on an earlier execution of the same benchmark performed on a consumer laptop GPU (NVIDIA GeForce RTX 4060 Laptop GPU, Windows, `cost_optimized` profile) and compared the two runs dataset by dataset under identical analysis code (Table S5).

Thirty-seven of the 44 datasets used byte-identical held-out partitions across the two runs, verified by split-signature hash; the other seven were re-split from random or target-quartile to scaffold splits in the A100 run and are therefore not directly comparable. On those 37 directly comparable datasets, the best single model (ensembles and fusion excluded, because the two runs built them under different protocols) changed the primary metric by a median of +0.50% — the A100 run was better on 22 datasets and the consumer-GPU run on 13. At this resolution the two hardware configurations are indistinguishable in accuracy.

The A100 bought capacity and throughput: a `full` rather than `cost-optimized` model profile, a longer Chemprop schedule (40 epochs, 3-model ensembles versus 15 epochs and single models), a longer Uni-Mol schedule with mixed precision, the addition of Uni-Mol V2 and a TensorFlow deep tabular backend, 32-way parallel feature selection, and a complete run in 112 hours rather than 155. Those changes are what produced the shifts in *which family wins* reported in main-text Section 3.2 — most notably the 3D pretrained family’s classification wins, which did not occur under the shorter schedule.

The practical implication for readers is favourable to the accessibility argument. A practitioner with a laptop GPU can expect the same accuracy from this pipeline as one with an A100; what they lose is the ability to train the most expensive backends to convergence, and the wall-clock headroom to search as widely. Neither the selection-protocol result nor the estimated placements in main-text Section 3.5 depend on datacentre hardware.

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